

Quaternion convolutional neural networks for hyperspectral image classification

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ABSTRACT

Quaternion convolutional neural networks (QCNNs) can capture quaternion features, which contain not only the contextual information among quaternion feature units but also utilize the quaternion algebra inside the quaternion feature units to express structural information. However, building efficient QCNNs for hyperspectral image (HSI) classification is a challenge due to the lack of methods to map real features into quaternion features and the missing key quaternion modules. This paper proposes methodologies to build QCNNs specifically for HSI classification and designs several key quaternion modules. Firstly, a novel quaternion feature encoder is presented to map real HSI features into quaternion features. Secondly, a paradigm is designed to conveniently modify classical convolutional neural networks to implement QCNNs for HSI classification. Thirdly, we propose a new separable quaternion convolutional neural network (SQNet) based on presented advanced quaternion modules, such as separable quaternion convolutions and quaternion attention mechanisms. In addition, the computational rules in quaternion neurons are improved to enhance the adaptability of QCNNs. Experiments conducted on four widely used HSI datasets indicate that the implemented QCNNs and the proposed SQNet achieve satisfactory results for HSI classification with limited training samples.

1. Introduction

Hyperspectral images (HSIs) comprise hundreds of narrow spectral bands from ultraviolet to infrared spectrum, usually represented as 3D data cubes. Each pixel contains a contiguous spectral signature, reflecting the physical and chemical properties of the objects (Gao et al., 2022b). In addition, the pixel and its surrounding neighbors provide abundant spatial contextual information to characterize the objects precisely and accurately (Shang et al., 2018). HSIs show great potential for observing objects or materials and have been widely applied in various fields (Shang et al., 2019), such as urban planning (Wang et al., 2022a), landscape characterization (Gao et al., 2022a), precision agriculture (Gao et al., 2021), and geological survey (van der Meer et al., 2012). As a crucial prerequisite for HSI interpretation, HSI classification is an important research topic and application.

In the early stage of HSI classification, sequential spectral features are employed to distinguish different objects (Paul et al., 2021). These methods always project high-dimensional spectral information into a low-dimensional feature space (Delalieux et al., 2012; Ho, 1998). The inherent correlation between hyperspectral pixels is neglected, so it is hard to achieve satisfactory classification. In order to extract the spatial features of HSIs, researchers have exploited several methods based on

manual and shallow feature representation, such as morphological-based (Benediktsson et al., 2005), statistical-based (Li et al., 2021b), and kernel function-based methods (Camps-Valls and Bruzzone, 2005).

Deep learning methods have become a hot topic in the remote sensing community due to their strong ability to extract features automatically (Feng et al., 2022). Recent studies have raised the effectiveness of HSI classification to a reasonably high level (Lu et al., 2020; Gao et al., 2019). For example, Safari et al. proposed a convolutional neural network (CNN) to learn joint spatial-spectral features over multiple scales (Safari et al., 2021). Compared with other deep learning methods, CNNs can more effectively mine local patterns for HSI classification. More recently, deep learning methods based on quaternions, which are hypercomplex numbers spanned by linearly independent basis $\{1, i, j, k\}$, have attracted much attention from researchers (Hirose and Yoshida, 2012). A quaternion can be regarded as a four-dimensional vector containing special computational rules. And the weights inside quaternion neurons can adequately capture structural features based on the rules of quaternion algebra (Parcollet et al., 2019a). A quaternion convolutional neural network (QCNN) combines a CNN and quaternions, exhibiting substantial feature extraction potential.

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Two challenges have hindered QCNNs for HSI classification. Firstly, the HSIs are naturally real data, requiring a special algorithm to map real features into quaternion features. The challenge of mapping the real features into quaternion features is not only to consider the spatial and spectral information but also to exploit the different components and the computation rules of quaternions effectively. Secondly, there is a lack of effective paradigms to build QCNNs for HSI classification. The modular model design facilitates the development of CNNs and makes it easy to build efficient models, so the other key to building QCNNs is to design advanced modules.

To address these challenges, we design a paradigm for building QCNNs and propose several key quaternion modules (i.e., the quaternion feature encoder, separable quaternion convolution, quaternion linear projection, quaternion channel attention mechanism, and quaternion spatial group attention mechanism). The proposed quaternion feature encoder efficiently maps real features into quaternion features. It obtains one real and three imaginary components by 3D average pooling and spatial-spectral axis decomposition, respectively. The classical CNNs are improved to implement QCNNs for HSI classification conveniently in the designed paradigm. Moreover, we refine the computational rules in QCNNs to enhance their adaptability. Based on these presented advanced quaternion modules, we propose a new separable quaternion convolutional network, named SQNet, based on the proposed quaternion feature encoder and other presented several key quaternion modules. Extensive experiments conducted on four HSI datasets demonstrate the superiority of our improved and proposed models.

The main contributions of this paper can be summarized as follows:

- We propose a novel quaternion feature encoder to map real HSI features into quaternion features so as to build QCNNs for HSI classification.
- We design a paradigm to modify classical CNNs to implement QCNNs for HSI classification conveniently.
- We present a new SQNet based on several proposed advanced quaternion modules to achieve high accuracy for HSI classification with limited training samples.

The rest of this paper is organized as follows. Section 2 introduces some related work. Section 3 describes the preliminaries. Section 4 presents the proposed method. Section 5 illustrates the experimental results. Finally, Section 6 concludes this paper.

2. Related work

2.1. Deep learning for HSI classification

Deep learning methods can automatically extract features and have been widely used in HSI classification (Meher, 2020; Luo et al., 2018). For instance, Liu et al. proposed a 2D-3D CNN with spatial-spectral multi-scale feature fusion, consisting of two information flows (Liu et al., 2021). The convolutional kernel benefits from its geometric properties and shows the advantages of extracting local spatial patterns.

The modular design has facilitated the development of deep learning, and CNNs have flexible scalabilities to keep their strong performance by introducing advanced modules. For example, the attentional mechanism is a hot topic in computer vision, and it has also been introduced to process HSI data as a plug-and-play module. Hang et al. proposed a CNN-based method with attention mechanisms to improve learning by obtaining a priori information from the HSI data (Hang et al., 2021). Mou et al. raised spectral-domain attention to alleviate the problem of spectral feature redundancy, inspired by the channel attention mechanism (Mou and Zhu, 2020). They introduced the gating mechanism into the attention module to selectively emphasize or suppress information and adaptively calibrate the spectral bands. Haut et al. embedded the attention mechanism into a backbone network with shortcut connections (Haut et al., 2019).

Although CNNs can efficiently utilize a priori kernel size, the geometric properties of convolutional kernels limit their ability to extract long-range dependencies (Feng et al., 2021). In contrast, the emerging self-attention mechanism is good at extracting long-range dependencies (Qing et al., 2021a). Some researchers exploited self-attention mechanisms to capture global contextual information (Qing et al., 2021b; Zhang et al., 2022). The self-attention mechanism-based methods require more training samples due to the lack of an inductive bias. From the feature perspective, the feature units extracted by CNNs are scalars, which limits the feature representation capability. Capsule networks utilize vectors as feature units and are good at representing diversified structural features. Paoletti et al. developed a CNN-based extension and redefined the capsules as specialized spatial-spectral units for HSI features (Paoletti et al., 2019b).

The spatial locations and spectral bands show different discriminatory properties in a 3D HSI data cube. Wang et al. designed a cross-attention mechanism to capture contextual information in orthogonal directions (Wang et al., 2022b). The cross-attention mechanism effectively combines the imaging principles of HSIs and effectively extracts the long-range dependence of various pixels. The neural architecture of spatial-spectral decomposition improves the efficiency of HSI feature extraction.

2.2. Quaternion neural networks

As a hypercomplex number, the quaternion has similar properties to a complex number. Recent studies on quaternions suggest that quaternion neural networks could have richer feature representation capabilities than real neural networks and complex neural networks (Liu et al., 2022). The motivation for quaternion methods is related to information theory and biology. Each unit of the quaternion feature can store more information. Parcollet et al. pointed out that quaternions allow the synthesis and processing of multidimensional features as entities to encode internal relationships and express external correlations (Parcollet et al., 2019b). In addition, there is a distinction in human neurons between synchronous and asynchronous properties, while synchronized neurons are more stable (Rakshit et al., 2018). The intrinsic structure of quaternions allows for more synchronization of quaternion neurons. Li et al. analyzed the relationship between synchronization and stability in quaternion neural networks (Li et al., 2021a).

The image processing methods based on quaternion neural networks are inspired by an intuitive and simple idea that the three imaginary components of quaternions are utilized to represent the three channels of RGB images separately. Parcollet et al. employed quaternion convolutional autoencoders to learn the dependencies between different channels of RGB images (Zhao et al., 2020). Yin et al. exploited QCNNs to extract additional feature representation and characterize the structure information of RGB images (Yin et al., 2019). Based on these early works, various modules of quaternion neural networks have been constructed, including quaternion convolutions, quaternion recurrent units, and quaternion capsule networks (Zhou et al., 2021, 2022).

Complex numbers and quaternions have been widely used in HSI classification. Peker et al. applied a complex neural network to HSI classification (Peker, 2021). They employed convolutional neural networks to encode the HSI as complex features and classify them with complex linear projections. Lin et al. enhanced HSIs based on two-dimensional quaternion singular spectrum analysis (Lin et al., 2021). Li et al. utilized principal component analysis to compress HSIs into three-channel images and represent them as quaternion features like RGB images (Li et al., 2019). This method inspired that the quaternion neural networks for RGB image processing are efficiently and directly transferable to HSI processing (Paoletti et al., 2019b). However, it compress too much spectral information that has a clear physical meaning and does not take advantage of the quaternion to represent spatial-spectral features.

In this paper, we propose several key quaternion modules and design a paradigm to implement QCNNs based on classical real CNNs.

The implemented QCNNs are improved feature extraction capabilities based on CNNs, which utilize the internal structure of quaternion feature units to represent coarse and fine-grained information simultaneously. Compared with the currently popular visual transformer methods, QCNNs combine long-range dependencies, and set the scale of convolutional kernels flexibly, like CNNs, to effectively utilize the a priori information. In addition, QCNNs are similar to the axis decomposition methods to extract features efficiently and express features by different components of quaternions, which makes the extracted features more comprehensive.

Compared with the existing quaternion neural network methods, we propose an efficient encoding method (quaternion feature encoder) for HSI data that reduces the information loss caused by dimension reduction to three-channel data. Moreover, we utilize the conjugate of quaternions to define the calculation of quaternion neurons, which improves the adaptability of QCNNs. In order to build high-performance QCNNs, we design a variety of advanced quaternion modules, thus building models flexibly. For instance, the raised quaternion attention mechanisms incorporate the fusion of information within the quaternion features based on the weighting of quaternion feature units (Chen et al., 2020). In particular, the presented quaternion spatial group attention mechanism considers coarse and fine-grained spatial information to increase the discriminative capability of models (Zhang et al., 2020). Based on these proposed modules, we build an efficient SQNet for high-accuracy HSI classification when the training samples are limited.

3. Preliminaries

3.1. Quaternion algebra

A quaternion $q = (r + xi + yj + zk) \in \mathbb{H}$ is a simple hypercomplex number with a real component ($r \in \mathbb{R}$) and three imaginary components ($x \in \mathbb{R}$, $y \in \mathbb{R}$, and $z \in \mathbb{R}$), where $\mathbb{H}$ is a quaternion domain. Its imaginary components represent the three orthogonal axes and are calculated according to the following rules.

$$\begin{aligned} i^2 = j^2 = k^2 = ijk = -1, \\ k = ij = -ji, j = ki = -ik, i = jk = -kj. \end{aligned} \quad (1)$$

The addition rule of two quaternions is defined as Eq. (2), where the real and imaginary components of two quaternions are added correspondingly. The Hamilton product is utilized to calculate the quaternion multiplication by Eq. (3), which does not obey the exchange law. Eq. (4) shows the conjugate q^* of a quaternion q . The modulo of q is calculated by Eq. (5).

$$q_1 + q_2 = (r_1 + r_2) + (x_1 + x_2)i + (y_1 + y_2)j + (z_1 + z_2)k, \quad (2)$$

$$\begin{aligned} q_1 \otimes q_2 &= (r_1 + x_1i + y_1j + z_1k)(r_2 + x_2i + y_2j + z_2k) \\ &= (r_1r_2 + r_1x_2i + r_1y_2j + r_1z_2k) + (x_1r_2i + x_1x_2i^2 + x_1y_2ij + x_1z_2ik) \\ &\quad + (y_1r_2j + y_1x_2ji + y_1y_2j^2 + y_1z_2jk) \\ &\quad + (z_1r_2k + z_1x_2ki + z_1y_2kj + z_1z_2k^2) \end{aligned} \quad (3)$$

$$\begin{aligned} &= (r_1r_2 - x_1x_2 - y_1y_2 - z_1z_2) + (x_1r_2 + r_1x_2 - z_1y_2 + y_1z_2)i \\ &\quad + (y_1r_2 + z_1x_2 + r_1y_2 - x_1z_2)j + (z_1r_2 - y_1x_2 + x_1y_2 + r_1z_2)k, \\ q^* &= r - xi - yj - zk, \end{aligned} \quad (4)$$

$$\|q\| = \sqrt{q \otimes q^*} = \sqrt{r^2 + x^2 + y^2 + z^2}. \quad (5)$$

In a quaternion q , r is utilized as a scalar to express the real component of the quaternion, and the imaginary components of the quaternion x , y , and z can be expressed by the vector $\mathbf{v} = [x, y, z]$. Thus, the quaternion q can be expressed by Eq. (6). The quaternion on polar coordinate is calculated by Eq. (7).

$$q = (r, \mathbf{v}), \quad (6)$$

$$q = \|q\|(\cos \theta + \mathbf{n} \sin \theta) = \|q\|e^{n\theta}, \cos \theta = \frac{r}{\|q\|}, \sin \theta = \frac{|\mathbf{v}|}{\|q\|}, \mathbf{n} = \frac{\mathbf{v}}{|\mathbf{v}|}, \quad (7)$$

$$\text{where } |\mathbf{v}| = \sqrt{x^2 + y^2 + z^2}.$$

3.2. Quaternion neurons

The quaternion neural network consists of quaternion neurons, and each neuron has input $x \in \mathbb{H}$, output $y \in \mathbb{H}$, weight $w \in \mathbb{H}$, and bias $b \in \mathbb{H}$ as quaternions in Eq. (8).

$$y = w \otimes x + b. \quad (8)$$

The backpropagation of the quaternion neural network separates the real and imaginary components of the quaternion. Eq. (9) shows the computational rule of gradient $\nabla_{\mathcal{L}}(w)$ for the real loss function $\mathcal{L}$ in backpropagation.

$$\nabla_{\mathcal{L}}(w) = \frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial w_r} + \frac{\partial \mathcal{L}}{\partial w_x}i + \frac{\partial \mathcal{L}}{\partial w_y}j + \frac{\partial \mathcal{L}}{\partial w_z}k, \quad (9)$$

where the quaternion weights $w = w_r + w_xi + w_yj + w_zk$.

3.3. Quaternion activation functions

Activation functions increase the nonlinear modeling property of neural networks. In this way, the quaternion activation function is an extension of the real activation function. Currently, there are two trains of thought in designing quaternion activation functions. The first one considers the quaternion as a whole, called fully quaternion activation functions. For instance, the hyperbolic tangent of a quaternion q is defined in Eq. (10).

$$\tanh(q) = \frac{e^{2q} - 1}{e^{2q} + 1}, e^q = e^r \left(\cos|\mathbf{v}| + \frac{\mathbf{v}}{|\mathbf{v}|} \sin|\mathbf{v}| \right). \quad (10)$$

The second one separates the quaternion components, named split quaternion activation functions. Taking the ReLU activation function as an example, Eq. (11) shows the split quaternion activation function of QReLU for a quaternion q .

$$\text{QReLU}(q) = \text{ReLU}(r) + \text{ReLU}(x)i + \text{ReLU}(y)j + \text{ReLU}(z)k, \quad (11)$$

where $\text{ReLU}(s) = \max\{0, s\}$, $s \in \mathbb{R}$.

4. Methods

4.1. Quaternion feature encoder

The proposed quaternion feature encoder aims to map real features into quaternion features so as to build QCNNs for HSI classification. As shown in Fig. 1(a), we consider that each pixel of HSI corresponds to a 1D spectral feature, and each wavelength band presents as a two-dimensional image. Moreover, pixels and their neighbors exhibit different spatial structures in the horizontal and vertical directions. Therefore, we can describe the spatial and spectral features from different perspectives.

Next, we illustrate the algorithm of the proposed quaternion feature decoder that maps real features into quaternion features. Given 3D HSI features $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, where C , H , and W are the numbers of channels, heights, and widths, respectively. Fig. 1(b) shows the HSI features are divided and mapped into quaternion components. The HSI features can be written as $\mathbf{F} = \{f_{c,h,w}\}_{c=1,h=1,w=1}^{C,H,W}$, where $f_{c,h,w} \in \mathbb{R}$. Let $\mathbf{f}_{h,w}^{(x)} = [f_{1,h,w}, f_{2,h,w}, \dots, f_{C,h,w}]^T$, $\mathbf{f}_{c,h}^{(y)} = [f_{c,h,1}, f_{c,h,2}, \dots, f_{c,h,W}]^T$, and $\mathbf{f}_{c,w}^{(z)} = [f_{c,1,w}, f_{c,2,w}, \dots, f_{c,H,w}]^T$ denote the decomposition of three axes, respectively. We utilize linear projections to map $\mathbf{f}_{h,w}^{(x)}$, $\mathbf{f}_{c,h}^{(y)}$, $\mathbf{f}_{c,w}^{(z)}$ into the three imaginary components of the quaternion features $\mathcal{G} = \mathcal{G}^{(r)} + \mathcal{G}^{(x)}i + \mathcal{G}^{(y)}j + \mathcal{G}^{(z)}k$, that is, $\mathbf{g}_{h,w}^{(x)} = \text{Linear}^{(x)}(\mathbf{f}_{h,w}^{(x)})$, $\mathbf{g}_{c,h}^{(y)} = \text{Linear}^{(y)}(\mathbf{f}_{c,h}^{(y)})$, and $\mathbf{g}_{c,w}^{(z)} = \text{Linear}^{(z)}(\mathbf{f}_{c,w}^{(z)})$ for $\mathcal{G}^{(x)} = \{\mathbf{g}_{h,w}^{(x)}\}_{h=1,w=1}^{H,W}$,

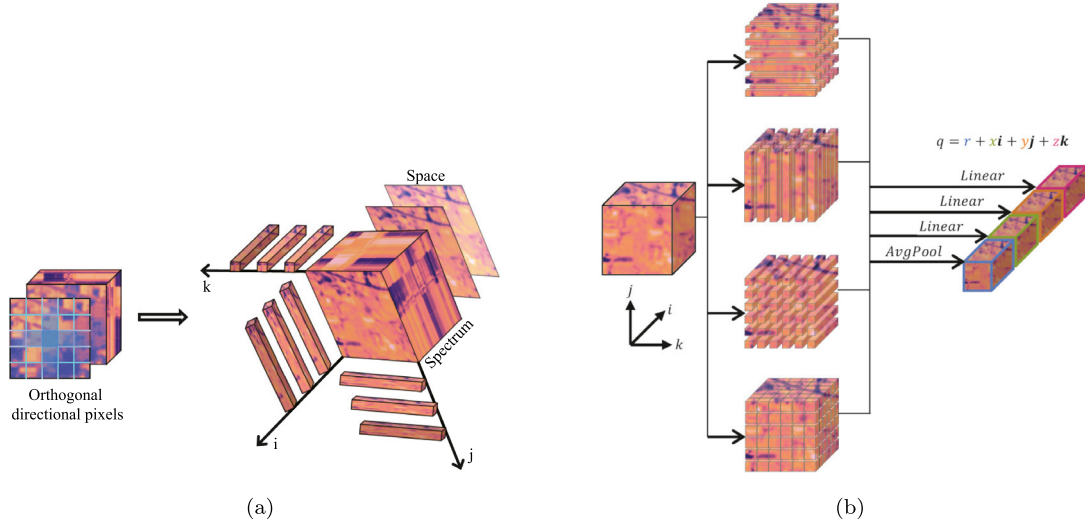


Fig. 1. Spatial-spectral features of an HSI and quaternion feature encoders. (a) The spectral and spatial structures (two spatial axes and a spectral axis) of an HSI data cube form three components whose orientation-dependent and context-dependent features are expressed in vectors. (b) Quaternion feature encoder. The real component of the quaternion features can be obtained by a 3D average pooling operation. The 3D cube is decomposed along the coordinate axes in different directions and then linearly projected into the imaginary components of the quaternion features, respectively.

$\mathcal{G}^{(y)} = \{g_{c,h}^{(y)}\}_{c=1,h=1}^{C,H}$, and $\mathcal{G}^{(z)} = \{g_{c,w}^{(z)}\}_{c=1,w=1}^{C,W}$, where $\mathcal{G} \in \mathbb{H}^{C \times H \times W}$, $g_{h,w}^{(x)} \in \mathbb{R}^C$, $g_{c,h}^{(y)} \in \mathbb{R}^W$, and $g_{c,w}^{(z)} \in \mathbb{R}^H$. The imaginary components of quaternion features record the fine-grained information of the object in different axes while expressing the long-range correlation of the pixels. In addition, we employ a 3D average pooling operation to obtain the real component $\mathcal{G}^{(r)}$, that is, $\mathcal{G}^{(r)} = \text{AvgPool}(\mathcal{F})$. Although quaternions are stored as four-dimensional vectors, they obey the algebraic rules of quaternions when calculated. The proposed quaternion feature encoder integrates the channel and spatial features to achieve a comprehensive and multiple-perspective feature representation. Algorithm 1 shows the algorithm of the proposed quaternion feature encoder.

Algorithm 1 Algorithm of the proposed quaternion feature encoder.

Input: Real feature $\mathcal{F} \in \mathbb{R}^{C \times H \times W}$.

Output: Quaternion features $\mathcal{G} \in \mathbb{H}^{C \times H \times W}$

- 1: Calculate the real component of the quaternion features:
 $\mathcal{G}^{(r)} = \text{AvgPool}(\mathcal{F})$.
 - 2: Calculate the three imaginary components of the quaternion features:
 $\mathcal{G}^{(x)} = \{\text{Linear}^{(x)}(f_{h,w}^{(x)})\}_{h=1,w=1}^{H,W}$
 $\mathcal{G}^{(y)} = \{\text{Linear}^{(y)}(f_{c,h}^{(y)})\}_{c=1,h=1}^{C,H}$
 $\mathcal{G}^{(z)} = \{\text{Linear}^{(z)}(f_{c,w}^{(z)})\}_{c=1,w=1}^{C,W}$
 - 3: Combine the real and imaginary components as the quaternion features:
 $\mathcal{G} = \mathcal{G}^{(r)} + \mathcal{G}^{(x)}i + \mathcal{G}^{(y)}j + \mathcal{G}^{(z)}k$
 - 4: **return** $\mathcal{G}$
-

4.2. Computation of quaternion convolutional neural networks

4.2.1. Computational rules for quaternion neurons

To increase the stability of the QCNNs, we propose a new computational rule, where the quaternion weights $\omega = \omega^{(r)} + \omega^{(x)}i + \omega^{(y)}j + \omega^{(z)}k$ is replaced by their conjugates $\omega^* = \omega^{(r)} - \omega^{(x)}i - \omega^{(y)}j - \omega^{(z)}k$, where $\omega \in \mathbb{H}$. The basic intuition behind this is to make the addition and subtraction calculations evenly distributed among the components. Let $s = s^{(r)} + s^{(x)}i + s^{(y)}j + s^{(z)}k$ denote the input data, where $s \in \mathbb{H}$. In this

way, we define the forward propagation by Eq. (12).

$$\begin{aligned} \omega^* \otimes s + \beta &= (\omega^{(r)}s^{(r)} + \omega^{(x)}s^{(x)} + \omega^{(y)}s^{(y)} + \omega^{(z)}s^{(z)}) \\ &\quad + (\omega^{(r)}s^{(x)} - \omega^{(x)}s^{(r)} - \omega^{(y)}s^{(z)} + \omega^{(z)}s^{(y)})i \\ &\quad + (\omega^{(r)}s^{(y)} + \omega^{(x)}s^{(z)} - \omega^{(y)}s^{(r)} - \omega^{(z)}s^{(x)})j \\ &\quad + (\omega^{(r)}s^{(z)} - \omega^{(x)}s^{(y)} + \omega^{(y)}s^{(x)} - \omega^{(z)}s^{(r)})k + \beta. \end{aligned} \quad (12)$$

where the bias $\beta \in \mathbb{H}$. The calculations in the later quaternion neurons use the conjugate weight calculation $\omega^* \otimes s + \beta$ by default. To facilitate the differentiation, we name the QCNN using original the computational rules for quaternion neurons $\omega \otimes s + \beta$ as a vanilla QCNN.

4.2.2. Quaternion convolution

The quaternion convolution preserves the characteristics of local connectivity and weight sharing while also incorporating the Hamilton product for quaternions. Moreover, we introduce the new computational rules for quaternion neurons into quaternion convolutions. Let $\mathcal{V}^* = \mathcal{V}^{(r)} - \mathcal{V}^{(x)}i - \mathcal{V}^{(y)}j - \mathcal{V}^{(z)}k$ denote the conjugate of quaternion convolutional kernels' weights, where $\mathcal{V} \in \mathbb{H}^{C_{\text{out}} \times C_{\text{in}} \times P \times P}$, C_{out} is the number of output features' channel, C_{in} is the number of input features' channel, and $P \times P$ is the size of quaternion convolutional kernel. Given the input feature $\mathcal{S} = \mathcal{S}^{(r)} + \mathcal{S}^{(x)}i + \mathcal{S}^{(y)}j + \mathcal{S}^{(z)}k$, the quaternion convolution can be written as $\mathcal{T} = \mathcal{V}^* * \mathcal{S}$, where $\mathcal{S} \in \mathbb{H}^{C_{\text{in}} \times H \times W}$ and $\mathcal{T} \in \mathbb{H}^{C_{\text{out}} \times H \times W}$. The quaternion convolution can be calculated by following equation:

$$T_{c_2,h,w} = \sum_{c_1=1}^{C_{\text{in}}} \sum_{\tau=-\lfloor \frac{P}{2} \rfloor}^{\lfloor \frac{P}{2} \rfloor} \sum_{\eta=-\lfloor \frac{P}{2} \rfloor}^{\lfloor \frac{P}{2} \rfloor} V_{c_2,c_1,\tau,\eta}^* \otimes S_{c_1,h+\tau,w+\eta}, \quad (13)$$

$$c_2 = 1, 2, \dots, C_{\text{out}}, \quad h = 1, 2, \dots, H, \quad w = 1, 2, \dots, W.$$

where $T_{c_2,h,w} \in \mathbb{H}$, $V_{c_2,c_1,\tau,\eta}^* \in \mathbb{H}$, and $S_{c_1,h+\tau,w+\eta} \in \mathbb{H}$ indicate values in the specific location of $\mathcal{T}$, $\mathcal{V}^*$, and $\mathcal{S}$, respectively. Values $S_{c_1,h+\tau,w+\eta}$ falling outside of $\mathcal{S}$ are padded by a constant value often defined as zero.

Similar to quaternion linear projections, the quaternion convolution can extract the spatial structure features effectively and fuse the information of various components by exchanging and sharing the convolutional operators in real and imaginary components. In addition, the quaternion convolution computes the residuals of various components, which alleviates the information bottleneck phenomenon. As shown in

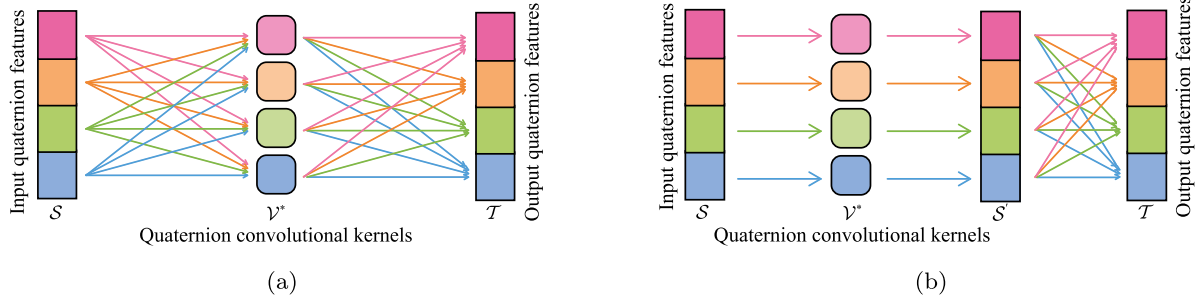


Fig. 2. Dense quaternion convolutions and Separable quaternion convolutions. (a) The quaternion convolutional kernels are densely acting on quaternion features in dense quaternion convolutions. (b) Separable quaternion convolutions are simplified as four sets of convolution operations and one set of mixing.

Fig. 2(a), the quaternion convolution can be expressed as the acting of the quaternion convolutional kernels' components to the quaternion features' components:

$$\begin{aligned} \mathcal{T} = \mathcal{V}^* * \mathcal{S} = & \left(\mathcal{V}^{(r)} * \mathcal{S}^{(r)} + \mathcal{V}^{(x)} * \mathcal{S}^{(x)} + \mathcal{V}^{(y)} * \mathcal{S}^{(y)} + \mathcal{V}^{(z)} * \mathcal{S}^{(z)} \right) \\ & + \left(\mathcal{V}^{(r)} * \mathcal{S}^{(x)} - \mathcal{V}^{(x)} * \mathcal{S}^{(r)} - \mathcal{V}^{(y)} * \mathcal{S}^{(z)} + \mathcal{V}^{(z)} * \mathcal{S}^{(y)} \right) i \\ & + \left(\mathcal{V}^{(r)} * \mathcal{S}^{(y)} + \mathcal{V}^{(x)} * \mathcal{S}^{(z)} - \mathcal{V}^{(y)} * \mathcal{S}^{(r)} - \mathcal{V}^{(z)} * \mathcal{S}^{(x)} \right) j \\ & + \left(\mathcal{V}^{(r)} * \mathcal{S}^{(z)} - \mathcal{V}^{(x)} * \mathcal{S}^{(y)} + \mathcal{V}^{(y)} * \mathcal{S}^{(x)} - \mathcal{V}^{(z)} * \mathcal{S}^{(r)} \right) k, \end{aligned} \quad (14)$$

where $*$ is the convolutional operation.

Obviously, the components of the quaternion convolutional kernel are densely acting on the components of the quaternion features. We name this quaternion convolution as a dense quaternion convolution. The computational complexity of the dense quaternion convolution is approximately $16HW C^2 P^2$, where C is the number of input and output channels, $H \times W$ is the size of the feature map, $P \times P$ is the kernel size.

4.2.3. Separable quaternion convolution

We propose separable quaternion convolution to improve the computational efficiency of quaternion convolution. From Eq. (14), we can observe that the dense quaternion convolution is computed approximately as sixteen banks of real convolutions and one set of mixing. The novel separable quaternion convolution simplifies the dense quaternion convolution to four sets of convolution operations and one set of mixing:

$$\begin{aligned} \mathcal{S}^{(r)} = \mathcal{V}^{(r)} * \mathcal{S}^{(r)}, \quad \mathcal{S}^{(x)} = \mathcal{V}^{(x)} * \mathcal{S}^{(x)}, \\ \mathcal{S}^{(y)} = \mathcal{V}^{(y)} * \mathcal{S}^{(y)}, \quad \mathcal{S}^{(z)} = \mathcal{V}^{(z)} * \mathcal{S}^{(z)}, \\ \mathcal{T} = \mathcal{V}^* \otimes \mathcal{S} = \left(\mathcal{S}^{(r)} + \mathcal{S}^{(x)} + \mathcal{S}^{(y)} + \mathcal{S}^{(z)} \right) \\ + \left(\mathcal{S}^{(x)} - \mathcal{S}^{(r)} - \mathcal{S}^{(z)} + \mathcal{S}^{(y)} \right) i \\ + \left(\mathcal{S}^{(y)} + \mathcal{S}^{(z)} - \mathcal{S}^{(r)} - \mathcal{S}^{(x)} \right) j + \left(\mathcal{S}^{(z)} - \mathcal{S}^{(y)} + \mathcal{S}^{(x)} - \mathcal{S}^{(r)} \right) k \\ = \left(\mathcal{V}^{(r)} * \mathcal{S}^{(r)} + \mathcal{V}^{(x)} * \mathcal{S}^{(x)} + \mathcal{V}^{(y)} * \mathcal{S}^{(y)} + \mathcal{V}^{(z)} * \mathcal{S}^{(z)} \right) \\ + \left(\mathcal{V}^{(x)} * \mathcal{S}^{(x)} - \mathcal{V}^{(r)} * \mathcal{S}^{(r)} - \mathcal{V}^{(z)} * \mathcal{S}^{(z)} + \mathcal{V}^{(y)} * \mathcal{S}^{(y)} \right) i \\ + \left(\mathcal{V}^{(y)} * \mathcal{S}^{(y)} + \mathcal{V}^{(z)} * \mathcal{S}^{(z)} - \mathcal{V}^{(r)} * \mathcal{S}^{(r)} - \mathcal{V}^{(x)} * \mathcal{S}^{(x)} \right) j \\ + \left(\mathcal{V}^{(z)} * \mathcal{S}^{(z)} - \mathcal{V}^{(y)} * \mathcal{S}^{(y)} + \mathcal{V}^{(x)} * \mathcal{S}^{(x)} - \mathcal{V}^{(r)} * \mathcal{S}^{(r)} \right) k, \end{aligned} \quad (15)$$

where $\otimes$ is the separable quaternion convolutional operation.

As shown in Fig. 2(b), the separable quaternion convolution can also be written as $\mathcal{T} = \mathbf{1}^* \otimes \mathcal{S}'$, where the quaternion weight tensor $\mathbf{1}^* = \mathbf{1} - \mathbf{1}i - \mathbf{1}j - \mathbf{1}k$ and $\otimes$ is the element-wise Hamilton product. The computational complexity of the separable quaternion convolution is approximately $4HW C^2 P^2$. As a result, the separable quaternion convolution based on the computational rules for quaternion neurons retains

the same number of parameters as the dense quaternion convolution and reduces the computational complexity by about 75%.

Based on the proposed quaternion feature encoder and the quaternion convolution, we design a paradigm that conveniently modifies classical CNNs to implement QCNNs for HSI classification. Specifically, we insert a quaternion feature encoder after the first real convolution of the CNN and then replace the subsequent real module with the quaternion module. Unlike the existing QCNNs (Yin et al., 2019; Zhou et al., 2021), the modified QCNNs have quaternion feature encoders that map real features into quaternion features, particularly designed for hyperspectral image classification. Moreover, the modified QCNNs have new computational rules and stronger adaptability.

4.3. Quaternion attention mechanism

The attention mechanism enables the model to highlight the discriminative information and suppress redundant information in features. As an embeddable module, the proposed quaternion mechanism computes the quaternion attention maps based on the conventional attention mechanism. The quaternion attention mechanism expresses each component's interdependence and promotes integration by activating the quaternion attention map and the quaternion algebra. The proposed quaternion attention mechanism includes the quaternion channel attention mechanism and the quaternion spatial group attention mechanism. Unlike the real attention mechanism, the proposed quaternion attention mechanism can not only emphasize the quaternion features that are beneficial to discriminate but also act inside the quaternion feature units to further integrate the components of quaternions.

4.3.1. Quaternion channel attention mechanism

We propose the quaternion channel attention mechanism to enhance the quaternion features. The quaternion channel attention mechanism is shown in Fig. 3(a). For the input quaternion feature $\mathcal{G} = \{g_{c,h,w}\}_{c=1,h=1,w=1}^{C,H,W}$, we employ the 2D global average pooling to compress the quaternion features by $\bar{\mathcal{G}} = \text{GlobalAvgPool}(\mathcal{G}) = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W g_{c,h,w}$, where $\mathcal{G} \in \mathbb{H}^{C \times H \times W}$ and $\bar{\mathcal{G}} \in \mathbb{H}^C$. Then, a quaternion linear projection (QLinear($\cdot$)) and an activation function (QReLU($\cdot$)) are utilized to squeeze the compressed feature $\tilde{\mathcal{U}} = \text{QReLU}(\text{QLinear}_{\text{com}}(\bar{\mathcal{G}}))$. After that, a quaternion linear projection is employed to excite the quaternion channel attention $\mathcal{U} = \text{QLinear}_{\text{exc}}(\tilde{\mathcal{U}})$, where $\mathcal{U} = [u_1, u_2, \dots, u_C]^T \in \mathbb{H}^C$. Finally, the quaternion channel attention $\mathcal{U}$ is weighted on the quaternion features $\hat{g}_{c,h,w} = u_c^* \otimes g_{c,h,w}$ ($c = 1, 2, \dots, C$, $h = 1, 2, \dots, H$, and $w = 1, 2, \dots, W$), where the output quaternion features $\hat{\mathcal{G}} = \{\hat{g}_{c,h,w}\}_{c=1,h=1,w=1}^{C,H,W}$ and $\hat{\mathcal{G}} \in \mathbb{H}^{C \times H \times W}$.

For quaternion feature units, the proposed quaternion channel attention mechanism not only emphasizes beneficial discriminative information and suppresses redundant information like the real channel attention mechanism (Hu et al., 2018) but also can act within the quaternion feature unit responding to structural information and achieve a weighted mixture of different components. In addition, the

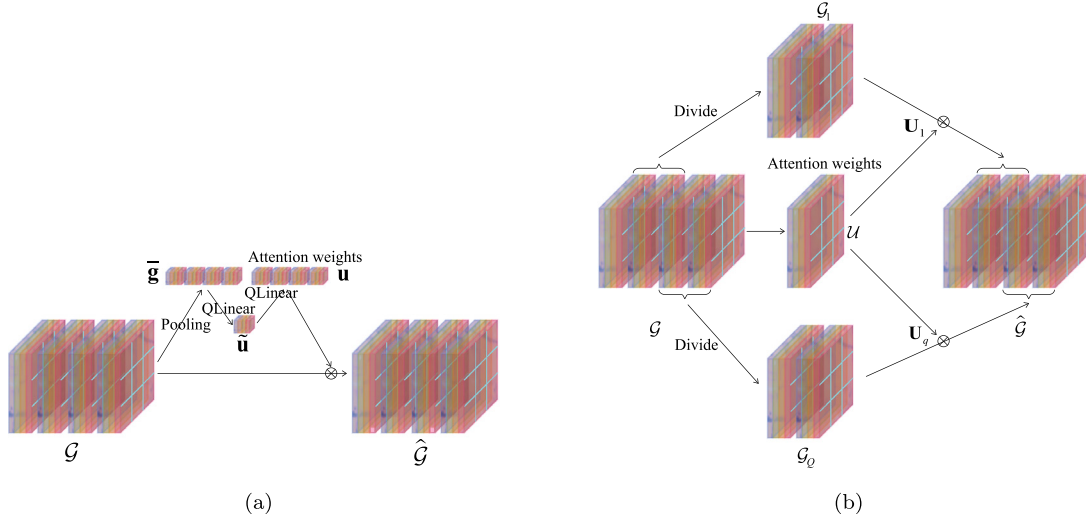


Fig. 3. Quaternion attention mechanism. (a) Quaternion channel attention mechanism. We utilize the 2D global average pooling to compress quaternion features. Then, the compressed quaternion features are employed to calculate the quaternion channel attention, which is implemented by quaternion linear projections and activation function. (b) Quaternion spatial group attention mechanism. The input features are divided into groups and utilized dense quaternion convolutions to calculate the weights for different groups.

backbone for feature extraction and the branches for computing attention in QCNNs are more explicit due to the non-exchangeability of quaternion multiplication. The details of the quaternion channel attention mechanism are shown in Algorithm 2.

Algorithm 2 Quaternion channel attention mechanism

Input: Input quaternion feature $\mathcal{G} \in \mathbb{H}^{C \times H \times W}$.
Output: Output quaternion feature $\hat{\mathcal{G}} \in \mathbb{H}^{C \times H \times W}$.
 1: Calculate the compressed quaternion features:
 $\bar{g} = \text{GlobalAvgPool}(\mathcal{G})$.
 2: Calculate the squeezed quaternion weights:
 $\tilde{u} = \text{QReLU}(\text{QLinear}_{\text{com}}(\bar{g}))$.
 3: Calculate the excited quaternion weights:
 $u = \text{QLinear}_{\text{exc}}(\tilde{u})$.
 4: Calculate the weighted quaternion features:
 $\forall g, c, h, w, \hat{g}_{c,h,w} = u_c^* \otimes g_{c,h,w}$.
 5: **return** $\hat{\mathcal{G}} = \{\hat{g}_{c,h,w}\}_{c=1,h=1,w=1}^{C,H,W}$.

4.3.2. Quaternion spatial group attention mechanism

We propose a quaternion spatial group attention mechanism to enhance the spatial structure and consider their coarse and fine grain features, as shown in Fig. 3(b). Let $\mathcal{G} \in \mathbb{H}^{C \times H \times W}$ denote the input quaternion features, where $\mathcal{G}$ can be written as $\mathcal{G} = [\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_A]$, $\mathcal{G}_a \in \mathbb{H}^{C_a \times H \times W}$, $C = \sum_{a=1}^A C_a$, and A is the number of channel groups. We utilize the dense quaternion convolution ($\text{QConv}(\cdot)$) and activation function ($\text{QReLU}(\cdot)$) to compress the quaternion features $\tilde{U} = \text{QConv}_{\text{com}}(\mathcal{G})$, where $\tilde{U} \in \mathbb{H}^{H \times W}$. Then, $\tilde{U}$ is utilized to excite the weights for different spatial groups by $U = \text{QConv}_{\text{exc}}(\tilde{U})$, where $U = [U_1, U_2, \dots, U_A]$, $U \in \mathbb{H}^{A \times H \times W}$ and $U_a \in \mathbb{H}^{H \times W}$ for $a = 1, 2, \dots, A$. The weight U_a can be written as $U_a = \{u_{a,h,w}\}_{h=1,w=1}^{H,W}$, where $u_{a,h,w} \in \mathbb{H}$ and let $g_{a,c,h,w}$ denote the element of $\mathcal{G}_a$ for $c = 1, 2, \dots, C$, $h = 1, 2, \dots, H$, and $w = 1, 2, \dots, W$. In the end, each element of the output features $\hat{\mathcal{G}}_a$ can be calculated by $\hat{g}_{a,c,h,w} = u_{a,h,w}^* \otimes g_{a,c,h,w}$, and the quaternion features in groups can be concatenated by $\hat{\mathcal{G}} = [\hat{\mathcal{G}}_1, \hat{\mathcal{G}}_2, \dots, \hat{\mathcal{G}}_A]$.

Unlike the existing real attention mechanisms, the proposed quaternion spatial group attention mechanism divides the quaternion features into different groups on the channel to handle different coarse and fine-grained spatial features. The proposed quaternion spatial group attention mechanism can maintain not only the diversified features but also facilitate the information exchange of the attention weights.

Algorithm 3 Quaternion spatial group attention mechanism

Input: Input quaternion feature $\mathcal{G} \in \mathbb{H}^{C \times H \times W}$.
Output: Output quaternion feature $\hat{\mathcal{G}} \in \mathbb{H}^{C \times H \times W}$.
 1: Split $\mathcal{G} = [\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_A]$.
 2: Calculate the compressed quaternion spatial weights:
 $\tilde{U} = \text{QConv}_{\text{com}}(\mathcal{G})$.
 3: Calculate the quaternion spatial weights for different groups:
 $U = \text{QConv}_{\text{exc}}(\tilde{U})$.
 4: Calculate the weighted quaternion features:
 $\forall a, c, h, w, \hat{g}_{a,c,h,w} = u_{a,h,w}^* \otimes g_{a,c,h,w}$.
 5: Concatenate quaternion features in different groups:
 $\hat{\mathcal{G}} = \text{Concat}(\hat{\mathcal{G}}_1, \hat{\mathcal{G}}_2, \dots, \hat{\mathcal{G}}_A)$.
 6: **return** $\hat{\mathcal{G}}$.

Algorithm 3 demonstrates the workflow of the quaternion spatial group attention mechanism.

4.4. Classifier

The classifier maps the quaternion feature vector to the corresponding category. Let $g = [g_1, g_2, \dots, g_B]^T \in \mathbb{H}^B$ denote the output features for classification, where B is the number of categories and g_b corresponds to the b th category for $b = 1, 2, \dots, B$. We compute the predictions $\hat{y} = [\hat{y}_1, \hat{y}_2, \dots, \hat{y}_B]^T \in \mathbb{R}^B$ for each category using the real component of g by the following equation:

$$\hat{y}_b = \text{softmax}(g_b^{(r)}) = \frac{\exp(g_b^{(r)})}{\sum_{a=1}^B \exp(g_a^{(r)})}, \quad b = 1, 2, \dots, B, \quad (16)$$

where $g_b = g_b^{(r)} + g_b^{(x)}i + g_b^{(y)}j + g_b^{(z)}k$.

4.5. The proposed sqnet

We present a new SQNet based on several proposed advanced quaternion modules for HSI classification, as shown in Fig. 4. The SQNet mainly includes a multi-scale separable quaternion convolution, separable quaternion convolutions, a quaternion channel attention mechanism, a quaternion spatial group attention mechanism, and a quaternion linear projection. The multi-scale separable quaternion convolution consists of four quaternion separable convolutions with kernel sizes of 1×1 , 3×3 , 5×5 , and 7×7 . The quaternion features

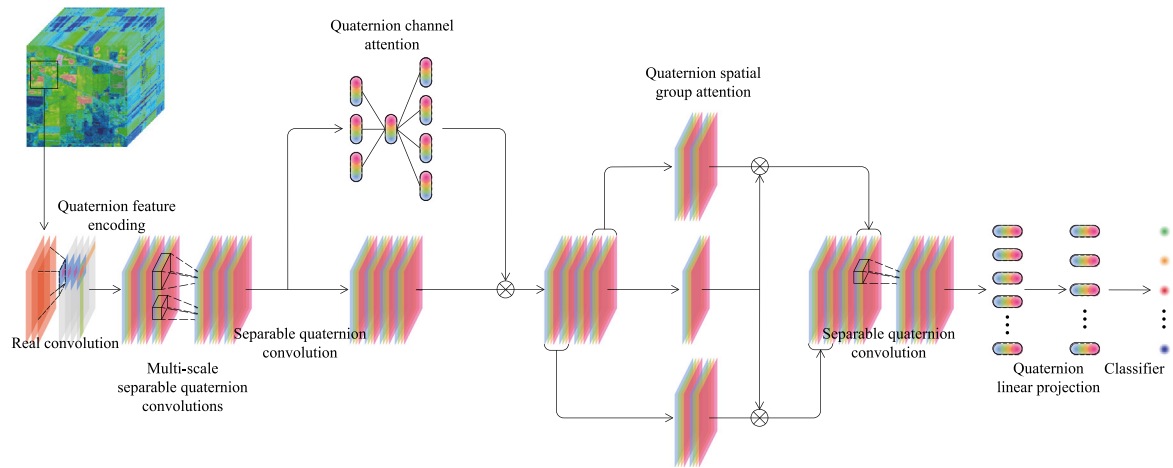


Fig. 4. Overview of the presented SQNet.

Table 1
Layers, operations, and hyperparameters of the proposed SQNet.

No.	Layers	Operations	Hyperparameters	
1	Real convolution	Real convolution	Kernel size = 3×3 Input channels = C_0	Padding = 1 Output channels = 64
2	Quaternion feature encoders	Real linear projection Real linear projection Real linear projection 3D average pooling	Input nodes = 64 Input nodes = 15 Input nodes = 15 Kernel size = 3×3	Output nodes = 64 Output nodes = 15 Output nodes = 15 Padding = 1
3	Activation function	QReLU	N/A	
4	Multi-scale separable quaternion convolution	Separable quaternion convolution	Kernel size = 1×1 Input channels = 64	Padding = 0 Output channels = 16
		Separable quaternion convolution	Kernel size = 3×3 Input channels = 64	Padding = 1 Output channels = 16
		Separable quaternion convolution	Kernel size = 5×5 Input channels = 64	Padding = 2 Output channels = 16
		Separable quaternion convolution	Kernel size = 7×7 Input channels = 64	Padding = 3 Output channels = 16
5	Activation function	QReLU	N/A	
6	Quaternion channel attention mechanism	Global average pooling Quaternion linear projection QReLU Quaternion linear projection	Input nodes = 64 Input nodes = 1 Input nodes = 1	N/A Output nodes = 1 Output nodes = 64
7	Separable quaternion convolution	Separable quaternion convolution	Kernel size = 3×3 Input channels = 64	Padding = 1 Output channels = 64
8	Activation function	QReLU	N/A	
9	Quaternion spatial group attention mechanism	Dense quaternion convolution QReLU Dense quaternion convolution	Kernel size = 3×3 Input channels = 64 Kernel size = 3×3 Input channels = 1	Padding = 1 Output channels = 1 Padding = 1 Output channels = 4
10	Separable quaternion convolution	Separable quaternion convolution	Kernel size = 3×3 Input channels = 64	Padding = 1 Output channels = 64
11	Activation function	QReLU	N/A	
12	Pooling	Global average pooling	N/A	
13	Quaternion linear projection	Quaternion linear projection	Input nodes = 64	Output nodes = B

extracted by different quaternion convolutions are concatenated on channels to represent coarse and fine-grained features. More details of the presented SQNet are listed in Table 1, where C_0 is the number of the input channels and B is the number of categories.

5. Experiments

In this section, the implemented QCNNs are validated on four widely used HSI datasets, and they are compared with the current state-of-the-art HSI classification methods. We visualize the quaternion features extracted by different HSI datasets. Finally, we analyze

the hyperparameter sensitivity and discuss the QCNNs from different perspective in ablation studies.

5.1. HSI datasets

5.1.1. Indian Pines HSI dataset

Indian Pines dataset¹ (Vane et al., 1993) is acquired by Airborne Visible Infra-Red Imaging Spectrometer (AVIRIS) instrument (see

¹ The data that support the findings of this study are openly available at https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes.

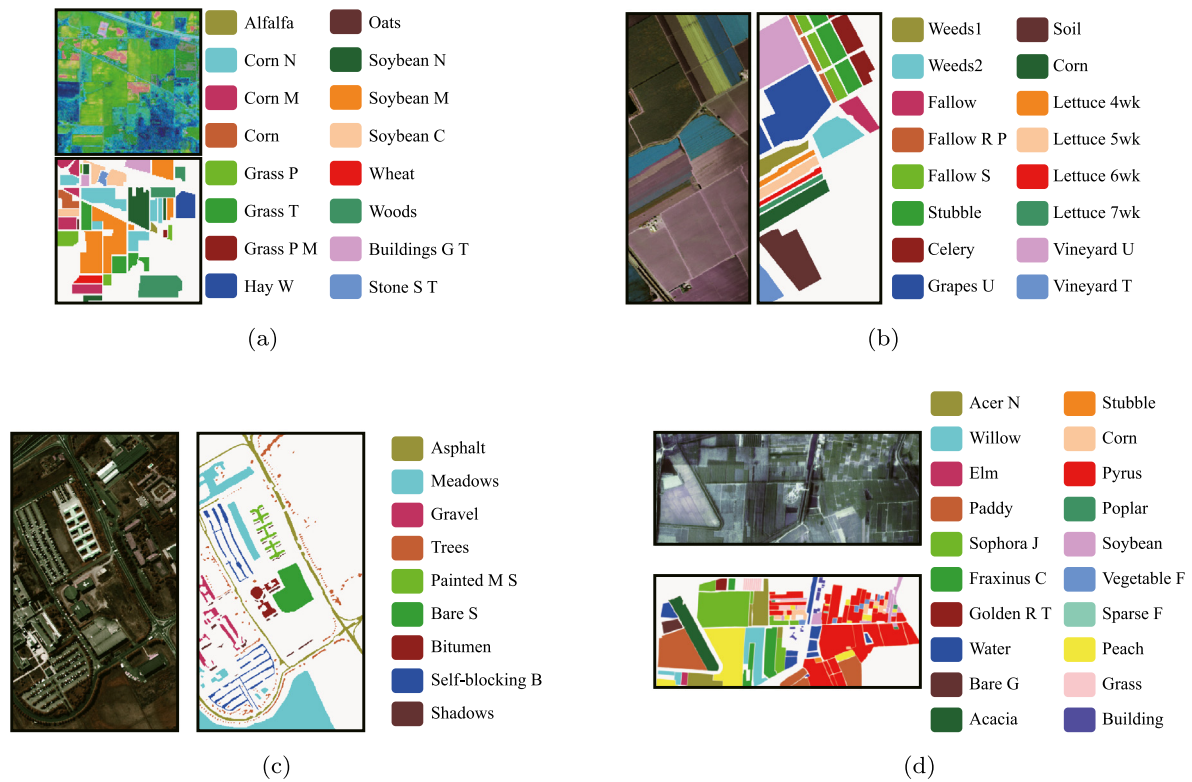


Fig. 5. The pseudocolor images and the corresponding ground truths of four HSI datasets. (a) Indian Pines. (b) Salinas. (c) Pavia University. (d) Xiongan.

Table 2

Groundtruth classes for the Indian Pines HSI dataset and their respective samples number.

No.	Classes	Samples	No.	Classes	Samples
1	Alfalfa	46	9	Oats	20
2	Corn N	1428	10	Soybean N	972
3	Corn M	830	11	Soybean M	2455
4	Corn	237	12	Soybean C	593
5	Grass P	483	13	Wheat	205
6	Grass T	730	14	Woods	1265
7	Grass P M	28	15	Buildings G T	386
8	Hay W	478	16	Stone S T	93
Total		10 249			

Fig. 5(a)). It is reduced to 200 spectral bands and has 145×145 pixels in space with a spatial resolution of 20 m per pixel. The dataset has 16 land cover classes with a mixture of features dominated by crops, grasslands, and woodlands. In addition, there is interference from crops in growth stages, bare soil, and planted crop residues, which bring challenges to the classification. Table 2 lists the number of samples for each class of the Indian Pines dataset.

5.1.2. Salinas HSI dataset

Salinas dataset² (Vane et al., 1993) is gathered by AVIRIS instrument, and it has 512×217 pixels and a spatial resolution of 3.7 m per pixel (see Fig. 5(b)). The water absorption bands are removed, and 204 spectral bands are kept in our experiments. The dataset has 16 land cover classes, among which the crop classes in different growth cycles are similar. Table 3 lists the number of samples for each class of Salinas datasets.

² The data that support the findings of this study are openly available at https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes.

Table 3

Groundtruth classes for the Salinas HSI dataset and their respective samples number.

No.	Classes	Samples	No.	Classes	Samples
1	Weeds1	2009	9	Soil	6203
2	Weeds2	3726	10	Corn	3278
3	Fallow	1976	11	Lettuce 4wk	1068
4	Fallow R P	1394	12	Lettuce 5wk	1927
5	Fallow S	2678	13	Lettuce 6wk	916
6	Stubble	3959	14	Lettuce 7wk	1070
7	Celery	3579	15	Vinyard U	7268
8	Grapes U	11 271	16	Vinyard T	1807
Total		54 129			

5.1.3. Pavia university HSI dataset

Pavia University dataset³ (Audebert et al., 2019) is captured by the ROSIS sensor (see Fig. 5(c)). It is reduced to 103 spectral bands with 610×340 pixels in space and a spatial resolution of 1.3 m per pixel. The dataset has nine land cover classes, mainly containing building roofs, grass, and asphalt. There are various isolated areas labeled in the dataset, which increases the fragmentation of features to some extent. Table 4 lists the number of samples for each class of Pavia University.

5.1.4. Xiongan new area HSI dataset

Xiongan New Area (Xiongan) dataset⁴ (Cen et al., 2020) is acquired in farming areas with various crop types over Hebei province, China, via a near-infrared imaging spectrometer on a UAV platform, of which the spectral range is 400–1000 nm, containing 256 bands (see Fig. 5(d)). It is built for precision crop classification with a spatial resolution of

³ The data that support the findings of this study are openly available at https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes.

⁴ The data that support the findings of this study are openly available at <http://doi.org/10.11834/jrs.20209065>.

Table 4

Groundtruth classes for the Pavia University HSI dataset and their respective samples number.

No.	Classes	Samples	No.	Classes	Samples
1	Asphalt	6631	6	Bare Soil	5029
2	Meadows	18 649	7	Bitumen	1330
3	Gravel	2099	8	Self-Blocking B	3682
4	Trees	3064	9	Shadows	947
5	Painted M S	1345			
Total		42 776			

Table 5

Groundtruth classes for the Xiongan HSI dataset and their respective samples number.

No.	Classes	Samples	No.	Classes	Samples
1	Acer N	225 647	11	Stubble	5612
2	Willow	180 766	12	Corn	59 165
3	Elm	15 353	13	Pyrus	1 026 513
4	Paddy	452 144	14	Poplar	7151
5	Sophora J	475 591	15	Soybean	91 072
6	Fraxinus C	169 342	16	Vegetable F	29 148
7	Golden R T	23 304	17	Sparse F	1496
8	Water	165 647	18	Peach	421 790
9	Bare G	38 409	19	Grass	65 514
10	Acacia	193 830	20	Building	29 616
Total		3 677 110			

0.5 m per pixel and contains 3750×1580 pixels. There are 3 677 110 labeled pixels in space.

Table 5 lists the number of samples for each class of Xiongan datasets.

5.2. Experimental settings

In this section, AlexNet (Krizhevsky et al., 2017), ResNet18 (He et al., 2016), Vgg11 (Simonyan and Zisserman, 2014), GhostNet (Paoletti et al., 2021), Capsule Networks (CapsNet) (Paoletti et al., 2019a), Vision Transformer (ViT) (Dosovitskiy et al., 2021) are considered as comparison methods with the proposed SQNet. We modify the three classical CNNs (AlexNet, ResNet18, and Vgg11) to build the QCNNs (QAlexNet, QResNet18, QVgg11), respectively. Specifically, the quaternion feature encoders are inserted into the three CNNs, and then the real convolutions are replaced with dense quaternion convolutions. More details of the QCNNs are described in the following.

QAlexNet: The first layer is composed of real convolutions whose kernel size is equal to 5×5 , and the output channel is set to 64. For the proposed quaternion feature encoder, the pooling kernel size is set to 3. The linear projections of the three imaginary components have 15 input nodes and 15 output nodes. The real convolutional layers and real linear projections are replaced with dense quaternion convolutions and quaternion linear projections, respectively.

QVgg11: The first two layers of QVgg11 consist of real convolutions and quaternion feature encoders, where the kernel size of convolutions and pooling operations is 3×3 . In addition, we replace the real convolutions and real linear projections with dense quaternion convolutions and quaternion linear projections, respectively.

QResNet18: QResNet18 comprises 18 layers, and it has the first real convolution with 3×3 kernel size. After that, the quaternion feature encoder maps real features into quaternion features. Finally, all subsequent real convolutions and real linear projections are also replaced with dense quaternion convolutions and quaternion linear projections, respectively.

SQNet: A novel quaternion convolution-based model is proposed with sequentially stacked a quaternion feature encoder, quaternion convolutions, and quaternion attention mechanisms, as listed in Table 1.

All comparison methods are implemented with the same experimental settings. The experiments are implemented based on PyTorch 1.9

and python 3.9 on a PC. An NVIDIA GeForce RTX 3090 with 24 GB of video memory is used. AdamW optimizer and cross-entropy loss function are utilized in the experiments. All models are trained with 200 epochs using Cross Entropy loss and AdamW optimizer. The initial learning rate is set to 0.0005 with a batch size of 64, and the spatial size for input patches is 15×15 . Each experiment is repeated five times, and the corresponding average values are reported.

5.3. HSI classification results

In the following experiments, we utilize the overall accuracy (OA), the average accuracy (AA), the Kappa coefficient (Kappa), and the training time (Time) for 200 epochs as criteria to evaluate the performance of all the compared methods. The simplest confidence interval is adopted using the mean (Mean) of the four HSI datasets in Tables 6, 7, 8, and 9. Since all parameters in the model participate in training and prediction, the training time and prediction time are positively correlated. Classification maps are visualized to compare different classification methods, as shown in Figs. 6, 7, 8, and 9.

From Table 6, Indian Pines has the least number of training samples among the four HSI datasets, and models are prone to overfitting problems. After introducing the quaternion algebra into the convolutional neural network, the most significant improvement in classification can be observed in the Indian Pines dataset. Specifically, the OA of QAlexNet, QResNet18, and QVgg11 are 2.01%, 0.05%, and 20.40% higher than AlexNet, ResNet18, and Vgg11, respectively. In particular, SQNet achieves the highest accuracy and is up to 91.68%. From Fig. 6, the misjudged pixels are mainly near the edges of each category region. The figure shows that the result of the QCNNs (QAlexNet, QResNet18, QVgg11, and SQNet) has significantly fewer pretzel-like and isolated misclassification points than that of the real CNNs. We believe that these phenomena are mainly due to the following two reasons. The first reason is that the quaternion feature encoder can extract long-range dependencies so that QCNNs (QAlexNet, QResNet18, QVgg11, and SQNet) will consider the properties of its neighboring image elements more when classifying the image elements. The second reason is that the QCNNs (QAlexNet, QResNet18, QVgg11, and SQNet) bind the four components as a whole, constraining the feature mapping of the QCNNs (QAlexNet, QResNet18, QVgg11, and SQNet) and reducing the fragmented features.

From Table 7 and Fig. 7, the misclassified pixels of the Salinas dataset are mainly concentrated in the categories of No. 8 (*Grapes U*) and No. 15 (*Vinyard U*). These two objects demonstrate homogeneous reflectance spectra and spatial structures. There is an increase in OA for QAlexNet, QResNet18, and QVgg11 by 1.6%, 1.19%, and 10.09% over AlexNet, ResNet18, and Vgg11, respectively. In particular, AlexNet performs well due to the high homogeneity of Salinas. The proposed SQNet also achieves the highest accuracy with an OA of 99.24%. Meanwhile, the training time of SQNet was significantly less than that of QResNet18 and QVgg11. This is because separable quaternion convolutions have less computational consumption than dense quaternion convolutions.

From Table 8, the object distribution of Pavia University dataset is fragmented. Therefore, the Pavia University dataset tends to rely on spectral information to distinguish between different categories. The OA of QAlexNet, QResNet18, and QVgg11 is 0.64%, 0.28%, and 1.79% higher than that of AlexNet, ResNet18, and Vgg11, respectively. From Fig. 8, the fragmented regions are mostly found in the classification map of real CNNs. Compared with the CapsNet, QCNNs (QAlexNet, QResNet18, QVgg11, and SQNet) not only express features through four-dimensional vectors, but also have the constraints of the computational rules for quaternion neurons on features.

From Table 9 and Fig. 9, the Xiongan dataset has high spatial resolution and many samples, which leads to severe spectral variability and spatial heterogeneity. As a result, it is a challenge to fuse spatial-spectral information while considering the fine spectral attribute and spatial geometry information. Comparing the results of

Table 6

Classification results for Indian Pines dataset using 2% of the labeled samples for training.

No.	AlexNet	QAlexNet	ResNet18	QResNet18	Vgg11	QVgg11	GhostNet	CapsNet	ViT	SQNet
1	99.28	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
2	84.10	87.32	90.36	88.21	68.88	88.89	69.26	87.74	83.92	92.02
3	78.15	76.95	80.36	81.85	27.47	76.75	55.86	82.17	71.41	93.25
4	97.18	98.03	97.47	98.59	96.20	98.45	90.15	96.62	93.11	96.77
5	89.65	83.44	91.30	81.09	46.17	87.16	73.57	87.72	83.30	89.51
6	82.06	88.50	89.36	91.00	78.77	88.91	82.92	87.26	88.59	89.68
7	100.00	100.00	100.00	100.00	97.62	100.00	100.00	100.00	100.00	100.00
8	95.47	98.05	97.84	94.14	88.35	95.95	87.94	86.61	96.03	99.65
9	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
10	85.32	83.78	86.49	88.07	65.09	84.53	67.94	77.02	76.65	87.65
11	89.14	91.99	91.78	92.07	83.35	93.26	89.41	94.47	89.51	91.87
12	73.97	84.54	84.93	89.88	15.74	92.13	61.38	86.90	75.71	85.39
13	98.37	99.19	95.77	95.93	94.63	97.40	94.31	96.58	87.97	95.77
14	97.47	97.92	98.16	98.13	97.29	99.34	96.31	97.42	98.34	98.84
15	62.44	65.89	58.72	64.16	20.12	63.30	53.02	60.54	63.56	69.86
16	97.49	97.49	92.11	95.70	97.85	96.06	98.21	93.91	94.98	98.21
OA (%)	86.69	88.70	89.68	89.73	69.46	89.86	78.72	88.47	85.50	91.68
AA (%)	89.38	90.82	90.92	91.18	73.60	91.38	82.52	89.69	87.69	93.03
Kappa (×100)	84.85	87.12	88.24	88.31	65.30	88.45	75.64	86.83	83.47	90.52
Time	4.27	6.10	6.00	32.07	4.97	19.23	14.60	12.57	11.73	10.23

Table 7

Classification results for Salinas dataset using 2% of the labeled samples for training.

No.	AlexNet	QAlexNet	ResNet18	QResNet18	Vgg11	QVgg11	GhostNet	CapsNet	ViT	SQNet
1	99.98	100.00	100.00	99.62	22.28	100.00	99.62	99.49	99.85	100.00
2	99.90	100.00	100.00	100.00	100.00	100.00	99.98	99.91	99.71	99.99
3	99.44	99.92	99.98	100.00	67.11	100.00	99.75	99.95	99.98	100.00
4	99.81	99.45	97.97	99.07	96.89	99.78	99.26	99.26	96.05	99.45
5	98.73	99.67	99.50	99.39	98.51	99.57	99.13	99.19	99.56	99.88
6	99.95	99.95	100.00	99.81	100.00	99.97	99.98	99.99	99.98	99.97
7	98.02	99.55	99.71	99.65	98.04	99.42	98.64	99.17	99.61	99.84
8	94.06	97.59	95.77	98.10	84.84	97.32	93.85	95.23	98.10	98.67
9	99.83	99.96	99.99	100.00	100.00	99.95	99.45	99.94	100.00	99.96
10	97.14	99.74	99.47	99.02	96.32	99.99	97.87	99.40	99.48	99.76
11	97.91	99.28	97.63	99.91	94.48	98.13	98.91	96.82	94.85	99.22
12	99.81	99.97	99.97	99.97	96.65	99.89	99.93	99.95	99.93	99.98
13	99.78	99.34	99.20	99.13	98.18	99.45	96.65	98.36	97.96	99.49
14	99.53	99.97	100.00	100.00	99.87	99.97	99.31	99.50	99.31	99.94
15	93.51	99.18	97.32	97.89	82.06	99.27	98.23	98.40	98.80	98.82
16	91.87	83.78	77.46	96.26	80.41	93.88	78.99	75.65	88.88	92.60
OA (%)	97.13	98.73	97.82	99.01	88.93	99.02	97.32	97.69	98.71	99.24
AA (%)	98.08	98.58	97.75	99.24	88.48	99.16	97.47	97.51	98.25	99.22
Kappa (×100)	96.81	98.59	97.57	98.90	87.65	98.90	97.01	97.43	98.56	99.15
Time	20.77	32.73	30.40	167.23	21.60	113.90	71.30	62.70	53.73	47.90

Table 8

Classification results for Pavia University dataset using 2% of the labeled samples for training.

No.	AlexNet	QAlexNet	ResNet18	QResNet18	Vgg11	QVgg11	GhostNet	CapsNet	ViT	SQNet
1	97.43	99.44	98.30	99.14	96.52	99.36	97.18	98.51	98.77	99.04
2	99.89	99.68	99.82	99.87	99.38	99.82	99.65	99.74	99.79	99.74
3	95.24	96.55	95.75	98.06	81.66	96.35	84.22	92.38	94.71	98.17
4	95.83	96.09	96.58	96.33	96.15	96.67	96.07	95.50	94.42	96.69
5	100.00	99.60	100.00	99.23	99.93	99.93	99.70	99.46	98.81	98.56
6	99.46	99.79	99.71	99.18	99.59	99.85	98.79	99.62	99.30	99.78
7	91.81	97.22	95.59	98.97	90.97	96.44	93.31	93.13	97.39	98.22
8	97.00	98.84	98.14	98.72	96.94	99.01	96.52	97.94	98.51	98.60
9	99.26	98.66	99.16	97.22	96.69	97.96	95.74	97.82	97.15	99.12
OA (%)	98.43	99.07	98.85	99.13	97.35	99.14	97.60	98.46	98.66	99.15
AA (%)	97.32	98.43	98.12	98.53	95.32	98.38	95.69	97.12	97.65	98.66
Kappa (×100)	97.91	98.77	98.48	98.85	96.49	98.86	96.82	97.96	98.23	98.87
Time	16.87	25.67	24.53	126.02	17.63	79.77	55.33	49.43	43.20	41.60

Table 9

Classification results for Xiongan dataset using 0.2% of the labeled samples for training.

No.	AlexNet	QAlexNet	ResNet18	QResNet18	Vgg11	QVgg11	GhostNet	CapsNet	ViT	SQNet
1	83.93	95.28	97.06	96.74	93.00	97.40	94.65	92.69	95.44	97.31
2	93.16	98.49	99.18	99.01	97.95	99.12	97.89	97.29	98.49	98.99
3	96.20	99.03	99.49	99.50	99.61	99.51	98.85	94.96	98.19	99.56
4	98.94	99.40	99.68	99.68	99.40	99.63	99.52	99.22	99.46	99.56
5	91.02	97.27	97.15	98.35	96.57	98.46	96.89	96.11	98.38	98.66
6	96.53	99.16	98.63	99.36	98.89	99.30	99.05	97.67	99.18	99.39
7	95.36	99.63	99.05	99.75	99.72	99.97	98.51	97.70	99.74	99.94
8	96.51	98.81	98.49	99.17	98.69	99.39	98.04	97.10	98.97	99.48
9	95.24	98.88	99.56	99.64	98.25	99.72	98.73	98.17	99.49	99.77
10	98.72	99.42	99.51	99.50	99.23	99.56	99.01	99.13	99.32	99.56
11	34.34	78.76	83.86	89.37	69.61	85.86	75.90	57.33	81.01	90.49
12	83.51	91.00	89.52	93.10	92.10	93.99	89.15	87.46	91.99	93.81
13	94.77	97.71	98.83	98.74	97.47	98.49	97.31	97.46	98.05	98.64
14	21.18	70.08	70.59	70.49	26.27	74.79	64.76	47.05	58.00	72.05
15	74.62	89.81	90.53	91.80	87.53	91.85	88.25	87.12	90.01	92.26
16	41.15	66.35	70.38	74.58	59.49	73.76	62.40	53.42	69.68	73.18
17	13.59	20.77	34.18	22.68	0.00	31.73	12.12	5.55	16.24	30.68
18	90.95	97.01	97.47	98.27	95.60	97.94	96.67	96.76	97.15	97.98
19	85.36	94.77	95.23	96.18	93.88	96.52	91.65	92.42	94.96	96.42
20	81.46	94.82	96.44	95.31	93.52	94.49	94.02	91.08	96.63	95.03
OA (%)	92.35	97.16	97.72	98.12	96.42	98.09	96.66	96.09	97.49	98.16
AA (%)	78.33	89.32	90.74	91.06	84.84	91.57	87.67	84.28	89.02	91.64
Kappa ($\times 100$)	91.10	96.71	97.36	97.82	95.84	97.79	96.12	95.46	97.08	97.87
Time	120.67	181.47	178.43	963.83	143.60	582.60	436.47	375.47	369.70	331.13

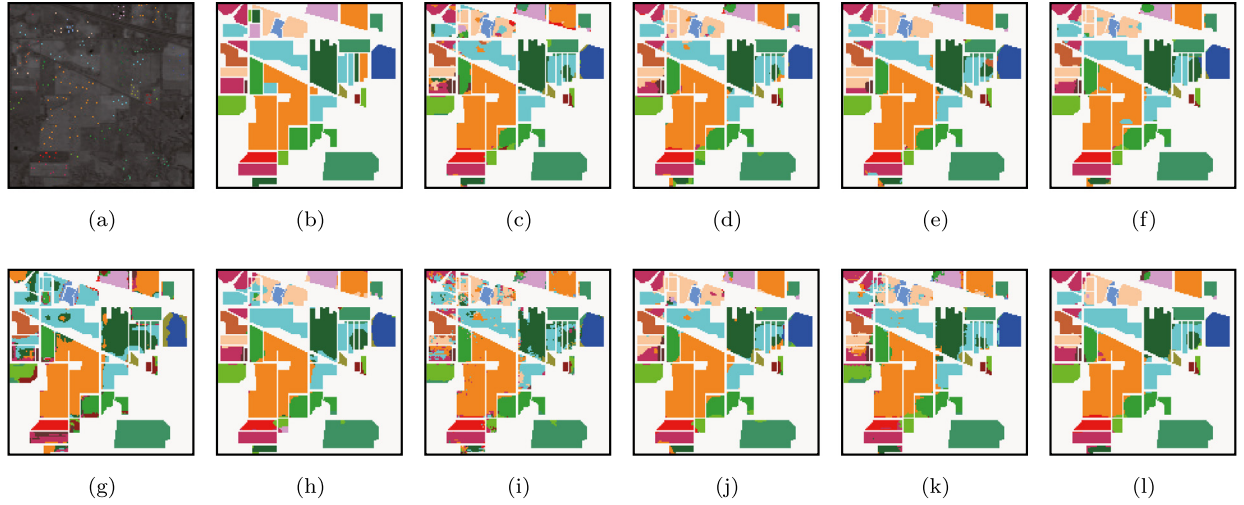


Fig. 6. Classification maps with 2% training samples for the Indian Pines dataset by different classifier. (a) Distribution of training samples. (b) Ground truth. (c) AlexNet (87.01%). (d) QAlexNet (89.43%). (e) ResNet18 (90.25%). (f) QResNet18 (90.73%). (g) Vgg11 (71.84%). (h) QVgg11 (90.94%). (i) GhostNet (82.19%). (j) CapsNet (90.15%). (k) ViT (86.21%). (l) SQNet (92.06%).

four real CNNs and their quaternion versions, the OA of QAlexNet, QResNet18, and QVgg11 are 4.81%, 0.4%, and 1.67% higher than that of AlexNet, ResNet18, and Vgg11, respectively. Specifically, for AlexNet and QAlexNet, the introduction of quaternions can represent complicated data structures effectively and increase the capacity of models. In the proposed quaternion feature encoder, the three imaginary components can remain on different dimensions of resolution separately, which is superior for extracting coarse and fine-grained features from high-resolution data.

5.4. Quaternion feature visualization

Firstly, taking the Indian Pines and Salinas datasets as examples, we visualize the features extracted by the quaternion feature encoders, as shown in Fig. 10. The overall quaternion features are concatenated by patches, where the real and imaginary components of the quaternion feature map highlight different objects, respectively. The encoded quaternion feature has better long-range dependence on the imaginary component. The output quaternion features of the encoder

are anisotropic due to the principle of imaginary component encoding. The quaternion components jointly extract texture structures for the different objects. Therefore, the quaternion features show better discriminative ability than the real features. Although it may be difficult to distinguish these features by a single real or imaginary component alone, it can obtain classification results based on quaternion features with high accuracy.

We visualize the features extracted by various networks by t-Distributed Stochastic Neighbor Embedding (t-SNE), as shown in Figs. 11, 12, 13, and 14. With the introduction of quaternion algebra in AlexNet, the feature extraction capability of QAlexNet is significantly enhanced. Taking the Xiongan dataset as an example (see Figs. 14(b) and 14(c)), QAlexNet has fewer overlapping regions of different categories of distributions. It indicates that the quaternion feature encoders and quaternion convolutions increase the discriminative and generalization of the model. The distribution of feature extracts by Vgg11 has a significant striped pattern (see Figs. 11(f), 12(f), 13(f), and 14(f)). The reason is that the last two linear projections of Vgg11 notably increase its fragmentation mapping capability, while the performance

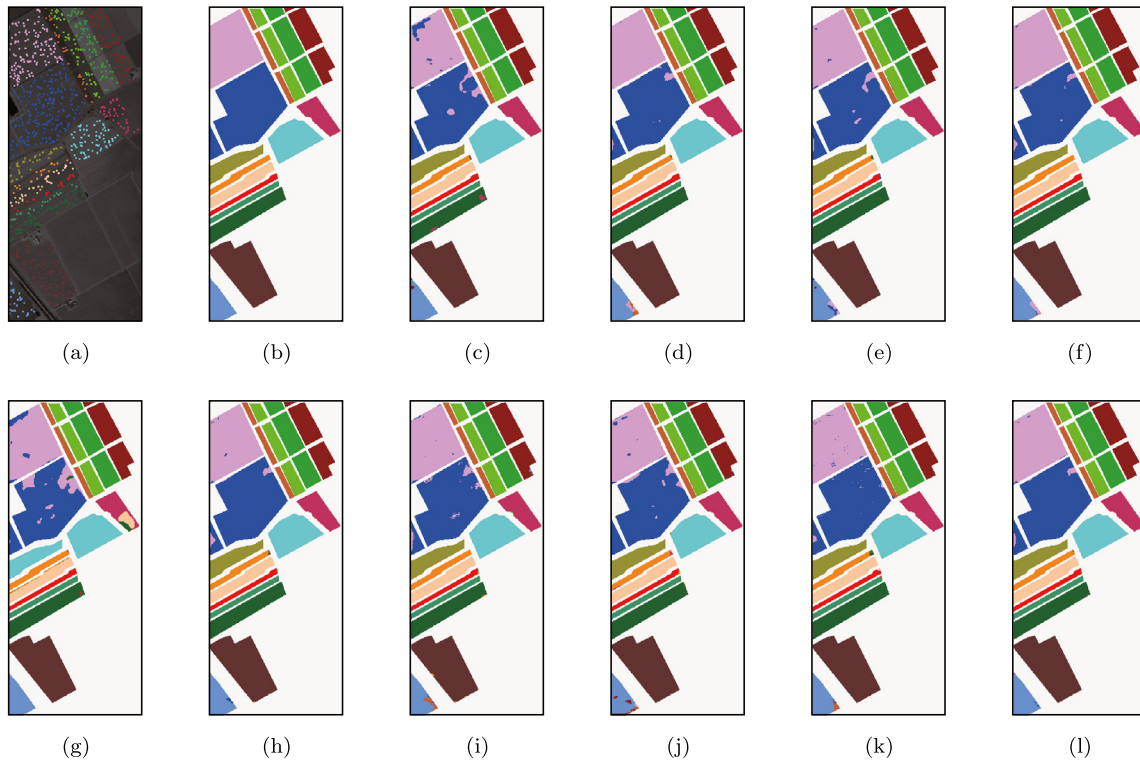


Fig. 7. Classification maps with 2% training samples for the Salinas dataset by different classifier. (a) Distribution of training samples. (b) Ground truth. (c) AlexNet (97.73%). (d) QAlexNet (98.97%). (e) ResNet18 (98.46%). (f) QResNet18 (99.08%). (g) Vgg11 (91.01%). (h) QVgg11 (99.13%). (i) GhostNet (98.06%). (j) CapsNet (97.97%). (k) ViT (99.00%). (l) SQNet (99.38%).

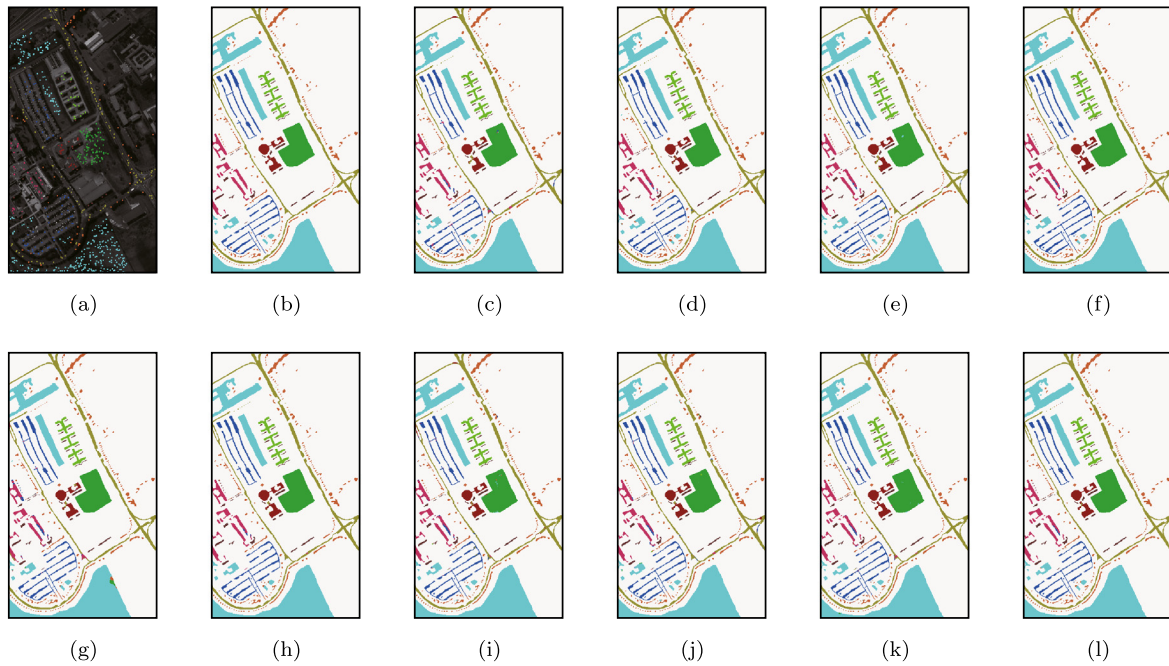


Fig. 8. Classification maps with 2% training samples for the Pavia University dataset by different classifier. (a) Distribution of training samples. (b) Ground truth. (c) AlexNet (98.56%). (d) QAlexNet (99.14%). (e) ResNet18 (99.00%). (f) QResNet18 (99.16%). (g) Vgg11 (97.81%). (h) QVgg11 (99.25%). (i) GhostNet (98.41%). (j) CapsNet (98.62%). (k) ViT (98.92%). (l) SQNet (99.28%).

shown by Vgg11 on the Pavia University dataset corroborates our view. In contrast, the distribution of feature extracts by QVgg11 (see Figs. 11(g), 12(g), 13(g), and 14(g)) is different from that of Vgg11. This phenomenon demonstrates that quaternion neurons constrain the fragmented mapping space. QResNet alleviates the overfitting problem of over ResNet. For the Salinas dataset, the area of confusion is severe

for ResNet18 (see Figs. 12(d)). As the accuracy of ResNet18 on different data is generally high, the improvement of QResNet18 is small. The distribution of feature extracts by SQNet shows a certain tendency of multiple clusters (see Figs. 11(k), 12(k), 13(k), and 14(k)). The representation of structure by quaternion features enables SQNet to distinguish objects with different structures in the same class. As a

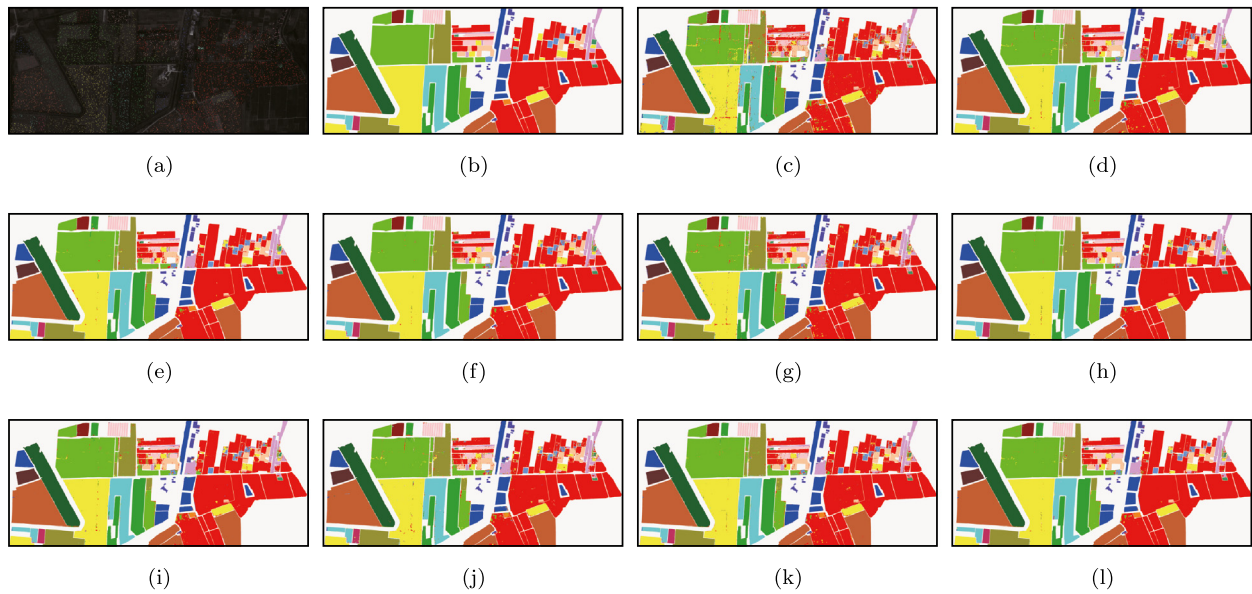


Fig. 9. Classification maps with 0.2% training samples for the Xiongan dataset by different classifier. (a) Distribution of training samples. (b) Ground truth. (c) AlexNet (92.53%). (d) QAlexNet (97.22%). (e) ResNet18 (98.01%). (f) **QResNet18 (98.22%)**. (g) Vgg11 (96.58%). (h) QVgg11 (98.18%). (i) GhostNet (96.74%). (j) CapsNet (96.31%). (k) ViT (97.54%). (l) **SQNet (98.22%)**.

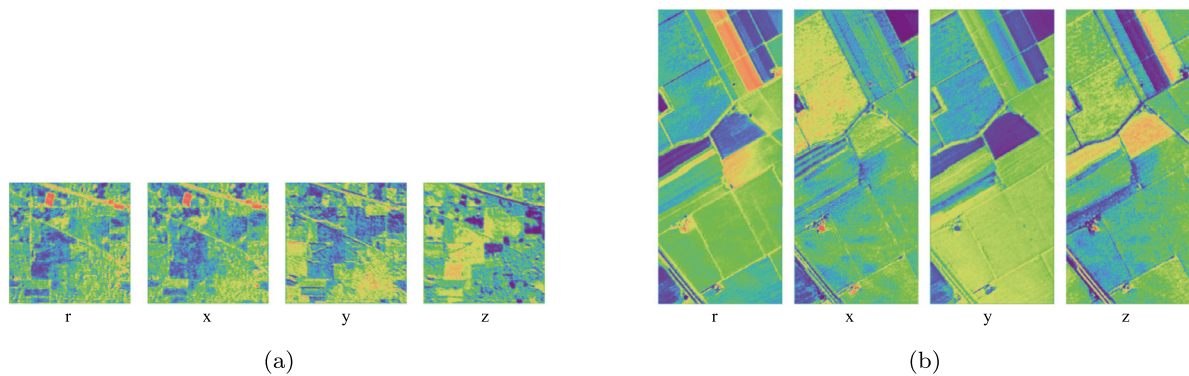


Fig. 10. The quaternion features for Indian Pines and Salinas datasets. After quaternion feature encoding, the feature maps are concatenated by multiple patches for Indian Pines and Salinas datasets. *r* indicates the real component of the quaternion. *x*, *y*, and *z* indicate the three imaginary components of the quaternion feature. (a) Indian Pines. (b) Salinas.

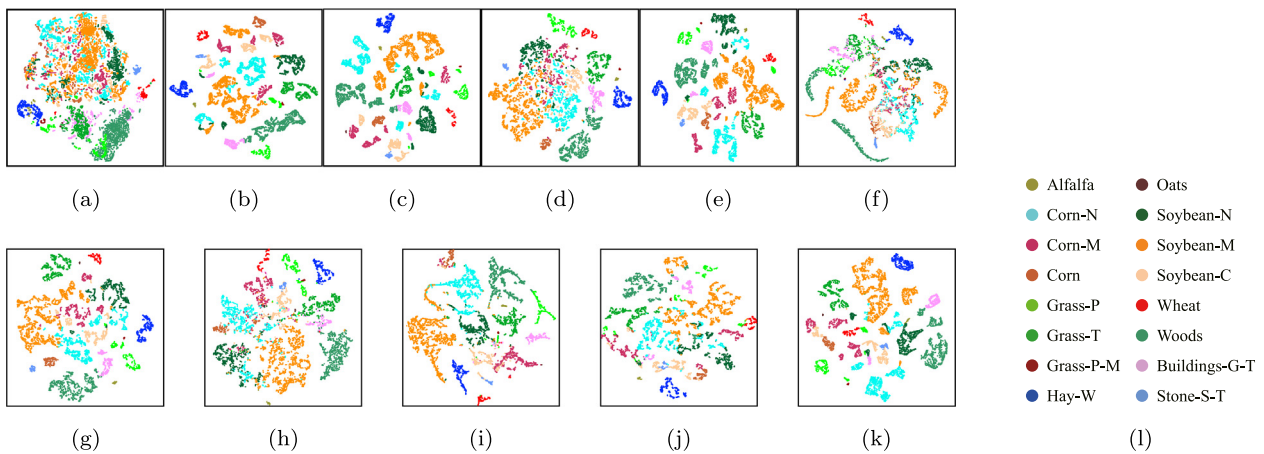


Fig. 11. Feature distribution visualization by t-SNE for Indian Pines dataset. (a) Raw data. (b) AlexNet. (c) QAlexNet. (d) ResNet18. (e) QResNet18. (f) Vgg11. (g) QVgg11. (h) GhostNet. (i) CapsNet. (j) ViT. (k) SQNet. (l) Legends.

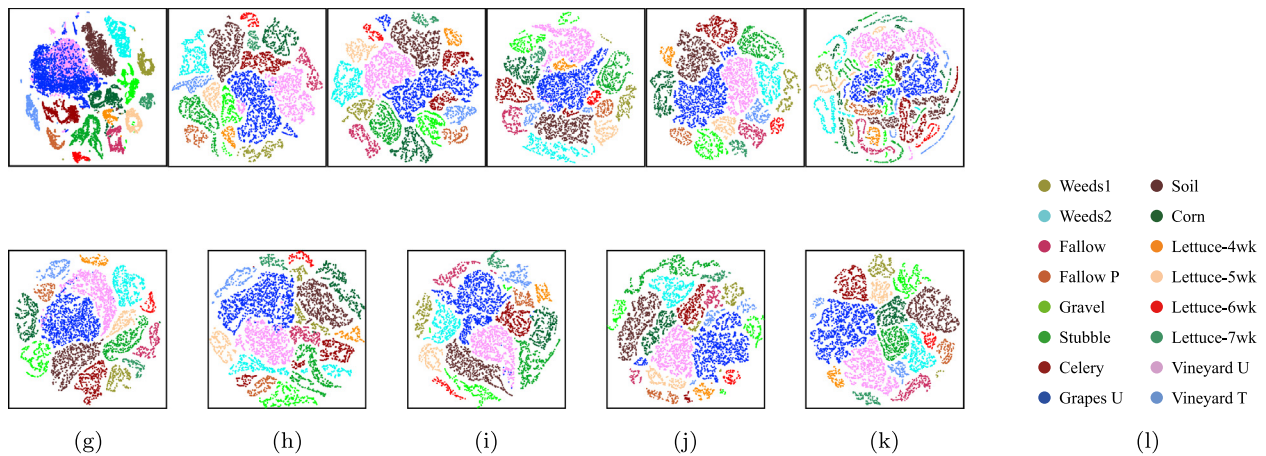


Fig. 12. Feature distribution visualization by t-SNE for Salinas dataset. (a) Raw data. (b) AlexNet. (c) QAlexNet. (d) ResNet18. (e) QResNet18. (f) Vgg11. (g) QVgg11. (h) GhostNet. (i) CapsNet. (j) ViT. (k) SQNet. (l) Legends.

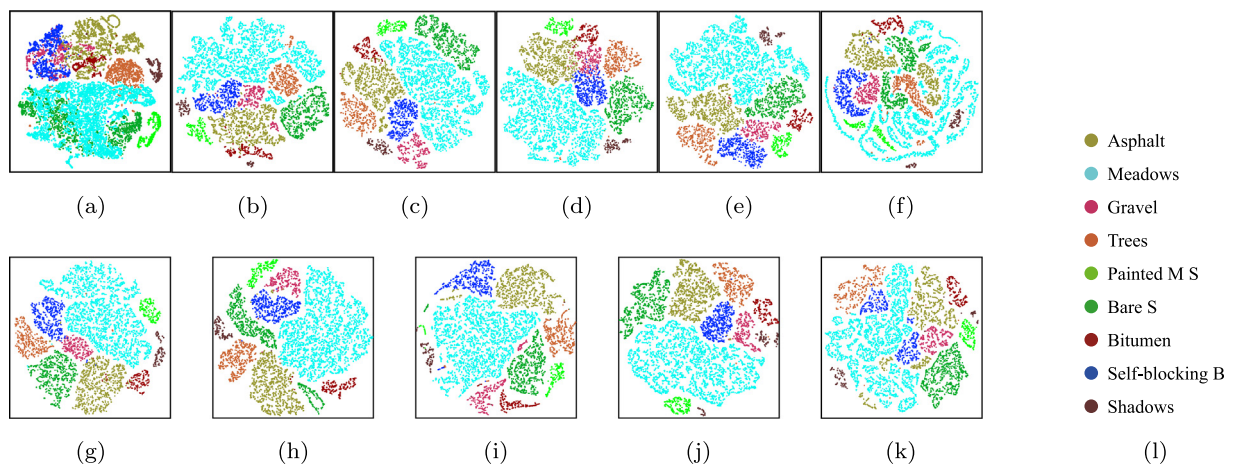


Fig. 13. Feature distribution visualization by t-SNE for Pavia University dataset. (a) Raw data. (b) AlexNet. (c) QAlexNet. (d) ResNet18. (e) QResNet18. (f) Vgg11. (g) QVgg11. (h) GhostNet. (i) CapsNet. (j) ViT. (k) SQNet. (l) Legends.

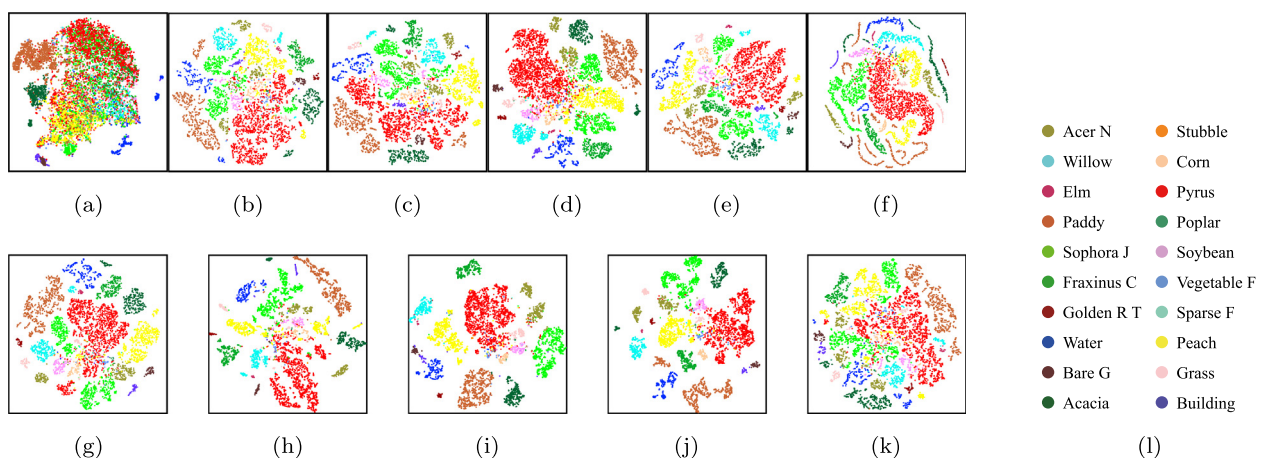


Fig. 14. Feature distribution visualization by t-SNE for Xiongan dataset. (a) Raw data. (b) AlexNet. (c) QAlexNet. (d) ResNet18. (e) QResNet18. (f) Vgg11. (g) QVgg11. (h) GhostNet. (i) CapsNet. (j) ViT. (k) SQNet. (l) Legends.

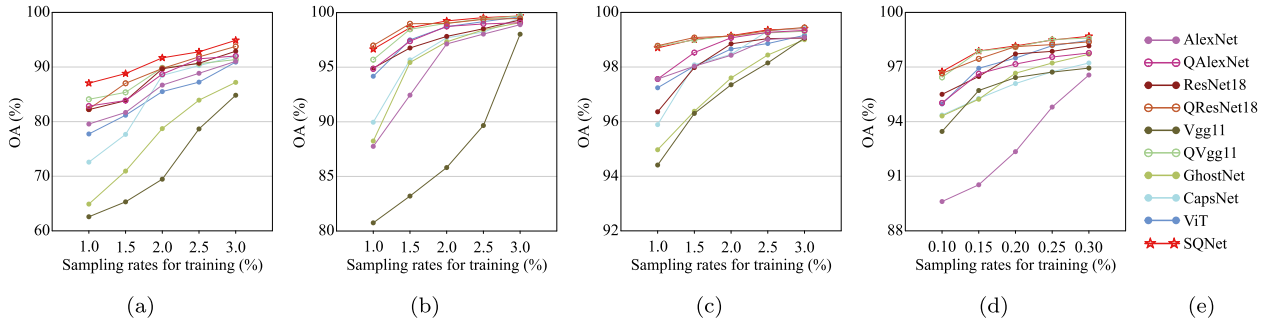


Fig. 15. The classification OA using different training samples. (a) Indian Pines. (b) Salinas. (c) Pavia. (d) Xiongan.

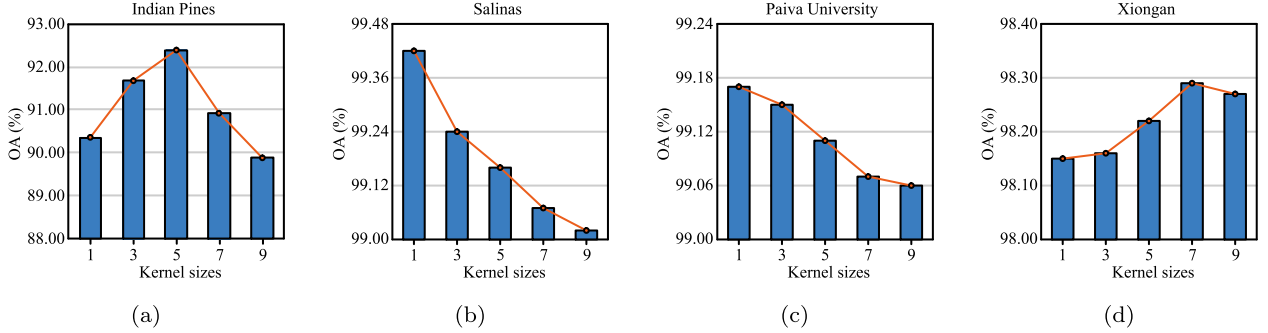


Fig. 16. The classification OA of the SQNet with different kernel sizes in the quaternion feature encoders. (a) Indian Pines. (b) Salinas. (c) Pavia. (d) Xiongan.

result, the representation of HSI features by SQNet is more relevant to practical application scenarios.

5.5. Experiments with different sampling rates

In this subsection, we analyze the influence of different sampling rates for training models. From Fig. 15, QCNNs (QAlexNet, QResNet18, QVgg11, and SQNet) have a greater advantage at lower sampling rates. As the sampling rate increases, the advantage of QCNNs gradually decreases. The Indian Pines dataset has the least number of labeled samples, so QCNNs still have a significant advantage using 3% of the labeled samples for training. For the Salinas dataset, the CapsNet and ViT achieve similar accuracies as the SQNet, because the feature units of CapsNet and ViT (capsules and tokens, respectively) are vectors. The feature units of the SQNet incorporate quaternion algebra based on four-dimensional vectors, so the SQNet has better accuracies when the training data set is limited.

5.6. Hyperparameter sensitivity

In this subsection, we analyze the hyperparameter sensitivity of the proposed SQNet. First, we compare the effect of the quaternion feature encoder's kernel size, as shown in Fig. 16. The kernel size is set to 1×1 , 3×3 , 5×5 , 7×7 , and 9×9 , respectively. The large kernel sizes imply that the real component of the quaternion feature is coarse-grained. It can be seen from the figure that the Salinas and Pavia University datasets have the highest accuracy at a kernel size of 1. This indicates that the indistinguishable parts of the Salinas and Pavia University datasets need to be extracted with more fine-grained features.

Then, we compare different widths of the proposed SQNet, where the width indicates the output channels from the first real convolution (see Table 1). As shown in Fig. 17, the Indian Pines and Xiongan datasets achieve the highest classification results at a width of 32. In contrast, the Salinas and Pavia University datasets achieve the highest accuracy at a width of 64. The smaller the SQNet width, the more spectral information is compressed by the real convolution. In addition, the larger the SQNet width is, the larger the capacity of the SQNet

is and the more likely it is to fall into overfitting. As a result, the classification accuracy of the SQNet on the four datasets decreases substantially when the width grows to 256.

5.7. Compared with vanilla QCNNs

To evaluate the computational rules in QCNNs, we compare the stability and robustness of two quaternion convolutions with different learning rates. Here, let $\mathcal{S}$ be the input features, $\mathcal{T}$ be the output features, and $\mathcal{V}$ be the weights of quaternion convolution kernels, we define the vanilla quaternion convolution as $\mathcal{T} = \mathcal{V} * \mathcal{S}$. In contrast, the quaternion convolution proposed in this paper is defined as $\mathcal{T} = \mathcal{V}^* * \mathcal{S}$, where $\mathcal{V}^*$ denote the conjugate of quaternion convolutional kernels' weights. Taking the QAlexNet and QResNet18 as examples, two quaternion convolutions are tested on the four HSI datasets. The proposed QAlexNet and QResNet18 employ the conjugate of quaternion weights to participate in the convolution. In contrast, quaternion weights in the vanilla QAlexNet (VQAlexNet) and vanilla QResNet18 (VQResNet18) are calculated directly. In this experiment, the performance of the two computational rules is compared in learning rates of 1×10^{-4} , 5×10^{-4} , 1×10^{-3} , 5×10^{-3} , 1×10^{-2} , and 5×10^{-2} . The number of epochs is set to 50 while the learning rate is constant. The rest of the settings are the same as in the previous experiments. The classification results with different learning rates are shown in Fig. 18. Both computational rules exhibit good performance under low learning rates.

With the increase in the learning rate, VQAlexNet and VQResNet18 tend to suffer from the vanishing gradient problem. Therefore, VQAlexNet and VQResNet18 need to learn at low learning rate conditions. In addition, the generalization of the VQAlexNet and VQResNet18 is weaker than that of QAlexNet and QResNet18, respectively. The experiments show that the new computational rules for quaternion neurons are more robust to the learning rate and have stronger generalizations than vanilla rules.

5.8. Effects of separable quaternion convolution

In this subsection, we designed experiments to verify the effects of separable quaternion convolutions. For the QAlexNet, QResNet18, and

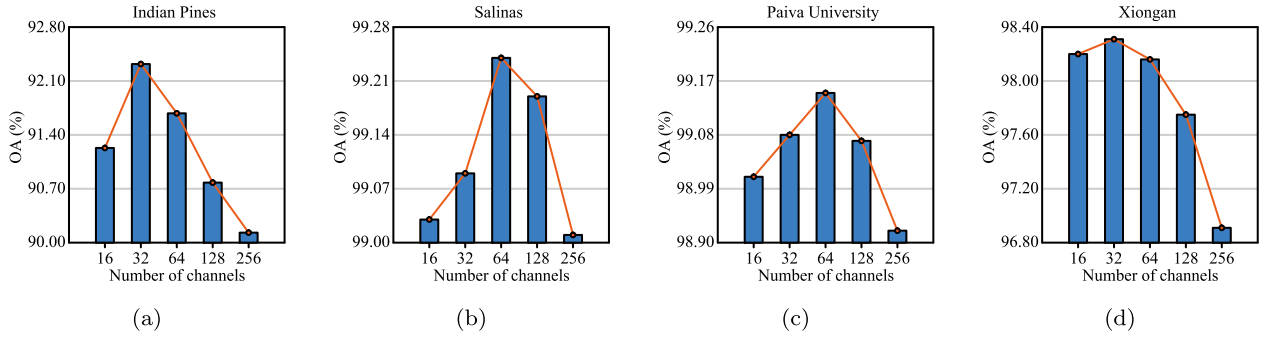


Fig. 17. The classification OA of the SQNet with different channels output from the first real convolution layer. (a) Indian Pines. (b) Salinas. (c) Pavia. (d) Xiongan.

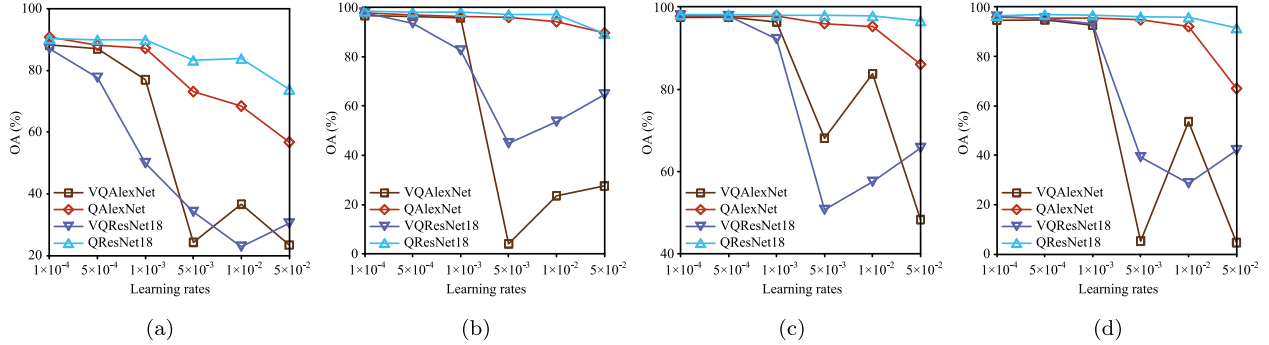


Fig. 18. The OA of two computational rules for quaternion neurons in 50 epochs with different learning rates. (a) Indian Pines. (b) Salinas. (c) Pavia. (d) Xiongan.

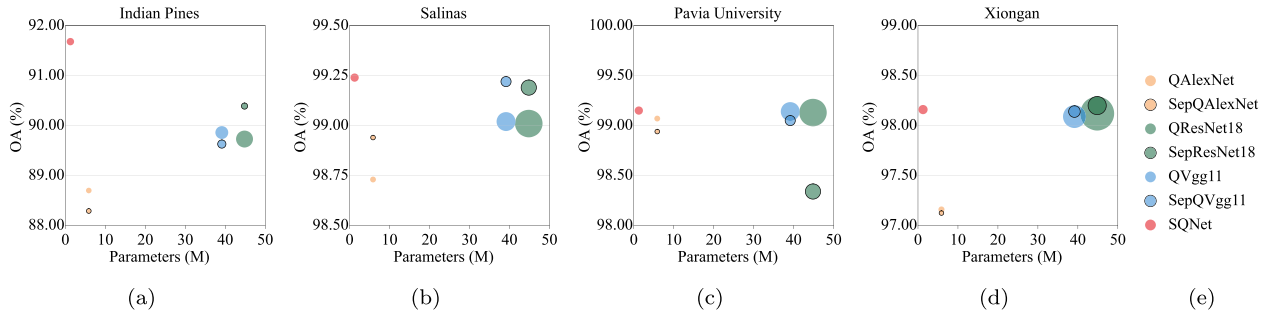


Fig. 19. The comparison of the classification OA and the number of model parameters for different QCNNs. The relative sizes of bubbles correspond to the running time. (a) Indian Pines. (b) Salinas. (c) Pavia. (d) Xiongan.

QVgg11, we utilize the dense quaternion convolutions for replacing the real convolution. In contrast, we design the SepQAlexNet, SepQResNet18, and SepQVgg11, which replace the real convolutions in AlexNet, ResNet18, and Vgg11 with separable quaternion convolution, respectively. Fig. 19 shows the comparison of the seven QCNNs (QAlexNet, SepQAlexNet, QResNet18, SepQResNet18, QVgg11, SepQVgg11, and SQNet). It can be seen from the figure that a separable quaternion convolution and a dense quaternion convolution have the same number of parameters, but the separable quaternion convolution reduces the training time. The comparison of accuracy shows that dense quaternion convolutions are more advantageous on the Pavia University dataset, and separable quaternion convolutions are better on both the Salinas and Xiongan datasets. In addition, the proposed SQNet has less number of parameters and better performance because the SQNet includes both dense quaternion convolutions (in the quaternion spatial group attention mechanism) and separable quaternion convolutions (see Table 1).

6. Conclusion and discussion

In this paper, we present advanced QCNNs for HSI classification. To this end, we design several effective quaternion modules and propose methodologies to build QCNNs specifically for HSI classification. Specifically, we present a novel quaternion feature encoder to map real HSI features into quaternion features, which combine long-distance dependence with coarse and fine-grained information. We refine the computational rules in QCNN so as to adapt to large learning rates, and the classical CNNs are improved to implement QCNNs effectively. Moreover, several key quaternion modules (separable quaternion convolutions, quaternion channel attention mechanisms, and quaternion spatial group mechanisms) are raised to enhance the QCNNs' capability of feature extraction. Based on these advanced quaternion modules, we propose an SQNet to efficiently achieve high-accuracy HSI classification, where separable quaternion convolution improves the computational efficiency of QCNNs. Comparison experiments, ablation experiments, and feature visualization demonstrate the effectiveness

of the proposed QCNNs. QCNNs can achieve high performance, outperforming representative deep learning methods when the training samples are limited.

In the future, more extensive tests will be conducted to apply the proposed network to other tasks. In addition, the computational efficiency of current QCNNs needs to be improved. We can further utilize quaternions to express the relationship between different structures. Moreover, improving the computational efficiency of QCNNs is also an important research direction.

CRedit authorship contribution statement

Heng Zhou: Methodology, Writing – original draft, Software. **Xin Zhang:** Conceptualization, Data curation. **Chunlei Zhang:** Supervision. **Qiaoyu Ma:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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